

Holocene vegetation change, climate variability, and human impacts on the Seki Basin (southwestern Türkiye): Evidence from a multiproxy approach

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ABSTRACT

This study presents a multiproxy palaeoenvironmental analysis of a 4.7 m sediment core from the Seki-Eceler Höyük (mound) archaeological site in southwestern Türkiye, providing a detailed reconstruction of vegetation and climate from 12000 yr BP to the present. Based on the first pollen record and geochemical proxies (μ XRF-measured elements and total organic and inorganic carbon), our findings reveal a clear transition from a cold and dry steppe environment during the Younger Dryas period to warmer and wetter conditions in the early Holocene. A major shift occurred during the early Holocene, characterised by a significant increase in arboreal pollen (*Quercus* and *Juniperus*) with *Pinus* and aquatic herbs, indicating a transition to wooded and wetter environments. This is supported by high Sr/Ca ratios and elevated μ -XRF concentrations of Ti, K and Fe, which suggest increased precipitation and hydrological activity. A sharp decline in arboreal pollen, which coincided with the 8.2 ka event, indicated a return to cold and arid conditions. This was followed by the Mid-Holocene Climatic Optimum (around 6000 yr BP), characterised by a significant increase in forest cover and biodiversity due to favourable, warmer and wetter conditions. During the late Holocene (4200 yr BP to present), the record shows a complex interplay of climatic shifts and increasing human impact. Although evidence of human presence in the region dates back to the Palaeolithic era, archaeological findings relating to grazing and pastoral activities are confined to the Neolithic era and later periods. This is consistent with the regional archaeological framework. While the 8.2 ka drought event may have influenced early subsistence strategies, clear signals of agro-pastoral practices only emerge after the onset of the Neolithic.

1. Introduction

Seki-Eceler Höyük, situated on the Seki Plateau within the Eşen (Xanthos) River basin in northern Lycia, has largely been overlooked in terms of its environmental and human history, with few archaeological and geomorphological investigations conducted (Aksoy and Köse, 2005; Momigliano et al., 2011a). Southwestern Anatolia (Türkiye), as part of the eastern Mediterranean region, has a Mediterranean climate and vegetation. The rapid transition in topography from coastal to high mountainous continental areas (Bakker et al., 2012) has resulted in multiple microenvironments with high biodiversity. These microregions

were settled by various cultures throughout the region's history (Eastwood et al., 1999; Médail and Quezél, 1999). Therefore, palaeoreconstruction studies involving palynology, palaeoclimatology and archaeology are needed to develop a comprehensive understanding of past human–climate–vegetation interactions at a microregional scale. Previous palynological studies in the vicinity of Seki-Eceler Höyük were conducted using cores extracted from the Sagalassos archaeological site in the ancient region of Pisidia, which is located northeast of Lycia. The sampling sites were established in archaeologically rich region (Vermeore et al., 1999, 2000, 2002; Kaniewski et al., 2007a, 2007b; Bakker et al., 2012, 2013). Although these cores are generally poorly dated, they

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cover a period from the early to the late Holocene onwards, and their results reveal the relationships among humans, vegetation, and climate in this region (Bakker et al., 2012, 2013).

When interpreting observed climate and vegetation changes, studies of these cores draw on comparisons with other regions, such as the Levant and the Aegean. Around the region, the early Holocene palaeoenvironment of Lake Gölhisar was investigated by analysing two sediment cores (Eastwood et al., 1998). Cores from Beyşehir I and II (van Zeist et al., 1975), Köyceğiz (van Zeist et al., 1975) and Lake Ova (Bottema and Woldring, 1984) can also be relatively securely assigned to the middle Holocene. Other cores, including those from Lakes Avlan and Karagöl in Elmalı (Bottema and Woldring, 1984), Lake Hoyran (van Zeist et al., 1975), and Güllük Bay (Biltekin et al., 2022), are assigned mainly to the late Holocene. In addition to palynological studies, other proxy studies conducted in southwestern Türkiye include oxygen stable isotope ($\delta^{18}\text{O}$) studies in Lake Gölhisar (Eastwood et al., 1998, 1999); advanced spaceborne thermal emission and reflection radiometer (ASTER) satellite imagery, charcoal analysis, cluster analysis (CA) and principal component analysis (PCA) of pollen percentages and concentrations in two cores from the Bereket area (Kaniewski et al., 2007a); μXRF studies in Isparta (Elmas et al., 2012); and in Lake Acıgöl, Denizli (Budakoğlu et al., 2015; Demory et al., 2020; Çaldırak et al., 2025); in Lake Köyceğiz (Akçer-Ön, 2017; Karadaş, 2022); and in the Dalyan Channel and Fethiye-Göcek Bay (Avşar et al., 2017), Lake Salda (Danladi et al., 2018), Denizli city (Oudeika et al., 2020), Tepe-Clazomenae Harbour (Izmir; Riddick et al. (2022), Yelten Marsh (Antalya, Şenkul et al., 2022), the Bozburun Peninsula (Gemici et al., 2022), the dried lakebed of Lake Avlan (Gozukara, 2022), Lake Gölçük (Danladi et al., 2023), Buldan Yayla Lake (Doğan et al., 2025), the Güllük Gulf (Biltekin et al., 2022; Duman et al., 2024); and magnetostratigraphy studies in the Soma Basin (Westerweel et al., 2020), and Lake Acıgöl were used for palaeoenvironmental reconstruction studies.

Numerous studies have focused on the reconstruction of Holocene climate patterns (e.g., Finné et al., 2011; Roberts et al., 2001, 2011a, 2019; Wanner et al., 2008, 2015; Holmgren et al., 2016; Jacobson et al., 2021). Some studies have also investigated past glacial intervals in southwestern Türkiye, for example, in the Bolkar Mountains (Çiner and Sankaya, 2017), the Lycia region (Sarıkaya and Çiner, 2015; Keserci et al., 2023), the Dedegöl Mountains (Köse et al., 2019), the Barla Mountains (Altınay et al., 2022), the Sandiras Mountains (Sarıkaya et al., 2008), the Akdağ Mountains (Sarıkaya et al., 2014) and the Geyikdağ Mountains (Sarıkaya et al., 2017). High-resolution, absolutely dated terrestrial records, particularly speleothem and dendrochronological datasets, provide valuable insights into regional hydroclimate variability and allow for accurate chronological correlations with global climate events. Speleothem records from caves such as Sofular, Dim, Öküzini and Kocain provide well-dated $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ time series reflecting changes in effective moisture and atmospheric circulation patterns in the eastern Mediterranean (Göktürk et al., 2011; Ünal-İmer et al., 2015, 2016; Rowe et al., 2020; Jacobson et al., 2021). These archives are valuable due to their high temporal resolution and robust U–Th chronologies, enabling the identification of rapid climatic events and their regional manifestations. In addition, dendrochronological studies from Anatolia provide records of hydroclimatic variability resolved to the year, offering an independent and highly sensitive proxy for reconstructing past precipitation and temperature fluctuations (Martinelli, 2004; Akkemik et al., 2019; Gmińska-Nowak et al., 2021; Yaman and Ekmen, 2023; Özarslan et al., 2025). Tree-ring data have been successfully employed to identify drought periods, extreme climatic events and long-term trends in climate variability in the eastern Mediterranean, thereby complementing lower-resolution sedimentary records. However, many Holocene climatic events and periods have not been clearly distinguished across much of southwestern Türkiye. This study aims to provide a high-resolution reconstruction of Holocene vegetation and climate dynamics in the highlands of central Lycia. This will help to fill a significant data gap in the region's

palaeoenvironmental history. To achieve this aim, our research addresses the following key questions: 1) How have the vegetation and climate of the Seki Plateau changed throughout the Holocene? 2) What is the local signature of major global and regional climatic events, and how does this signature compare with other records from southwestern Türkiye? 3) When and how did human land use begin to significantly alter the local landscape of vegetation? 4) What insights can we gain from the palaeoenvironmental record of Seki-Eceler Höyük regarding past human-environment interactions in the highlands of central Lycia?

1.1. Study area

The Seki Plateau ($36^{\circ}48'52.1''\text{N}$, $29^{\circ}38'13.1''\text{E}$; Fig. 1) is located in Muğla in southwestern Türkiye. It is a basin within the Lycian ophiolitic mélange (Danielian et al., 2006), and is surrounded by Lycian nappes (Fig. 2, Collins and Robertson, 1998; Danielian et al., 2006; Alçiçek et al., 2015; Keserci et al., 2023). It is bordered by the Akdağlar range to the west, and the Boncuk Dağları range to the northeast (Aksoy and Köse, 2005). As part of the wider eastern Mediterranean region, it experiences a Mediterranean climate characterised by mild, wet winters and long, hot summers (Özhatay et al., 2013; Lionello et al., 2023; Tüfekcioğlu and Tavşanoğlu, 2024). Wind patterns and climate are affected by pressure patterns such as the North Atlantic Oscillation (NAO) and the Mediterranean Oscillation (MO) (Cenk and Turgay, 2020; Criado-Aldeanueva and Soto-Navarro, 2020; Rowe et al., 2020). Located at an altitude of 1100 m, the Seki Basin has a cooler, drier climate than coastal regions do, with an annual rainfall of 571.2 mm and an average temperature of 12.7°C (Bozyiğit, 1997). Its elevated location is within the Oro-Mediterranean vegetation belt, which differs from coastal evergreen vegetation (Atalay et al., 2014; Roberts et al., 2011a, 2019). The area's high biodiversity and varied vegetation are attributed to its complex topography and the convergence of different climate regimes (van Zeist et al., 1975; Özhatay et al., 2013; Médail, 2022). The Seki Basin has a long history of human activity, serving as a seasonal highland pasture (yayla) for transhumant communities and as a key transportation corridor (Momigliano et al., 2011a). The most notable prehistoric settlement in the basin is Eceler Höyük. This site has been occupied since the late Neolithic/early Chalcolithic period and provides valuable archaeological data on the history of ancient Lycia (Aksoy and Köse, 2005; Momigliano et al., 2011a, 2011b, Momigliano, 2013; İşkan and Dündar, 2016).

2. Materials and methods

2.1. Coring studies

The JARK14–04 sediment core was recovered via a hydraulic vehicle-mounted rig adjacent to Seki-Eceler Höyük in southwestern Türkiye. The rig penetrated to a depth of 5.5 m, recovering 4.7 m of sediment. The sampling site was chosen in consultation with archaeologists to ensure that it was outside the protected zones.

2.2. Chronology

Ten bulk sediment samples from core JARK14–04 were sent to the Poznań Radiocarbon Laboratory for accelerator mass spectrometry (AMS) radiocarbon dating (Table 1). In this study, we revised the age-depth model previously carried out by Hashemi et al., 2026. In Table 1, TOC refers to untreated bulk sediment samples containing total organic carbon, as opposed to isolated terrestrial macrofossils. As they may incorporate mixed carbon sources, including reworked older carbon and younger rootlet contamination, bulk sediment samples can lead to anomalous radiocarbon ages. In addition, the potential impact of hard water (i.e. reservoir water) cannot be ruled out in this lacustrine system, particularly given the possible contribution of dissolved inorganic carbon derived from carbonate weathering within the catchment area. Such

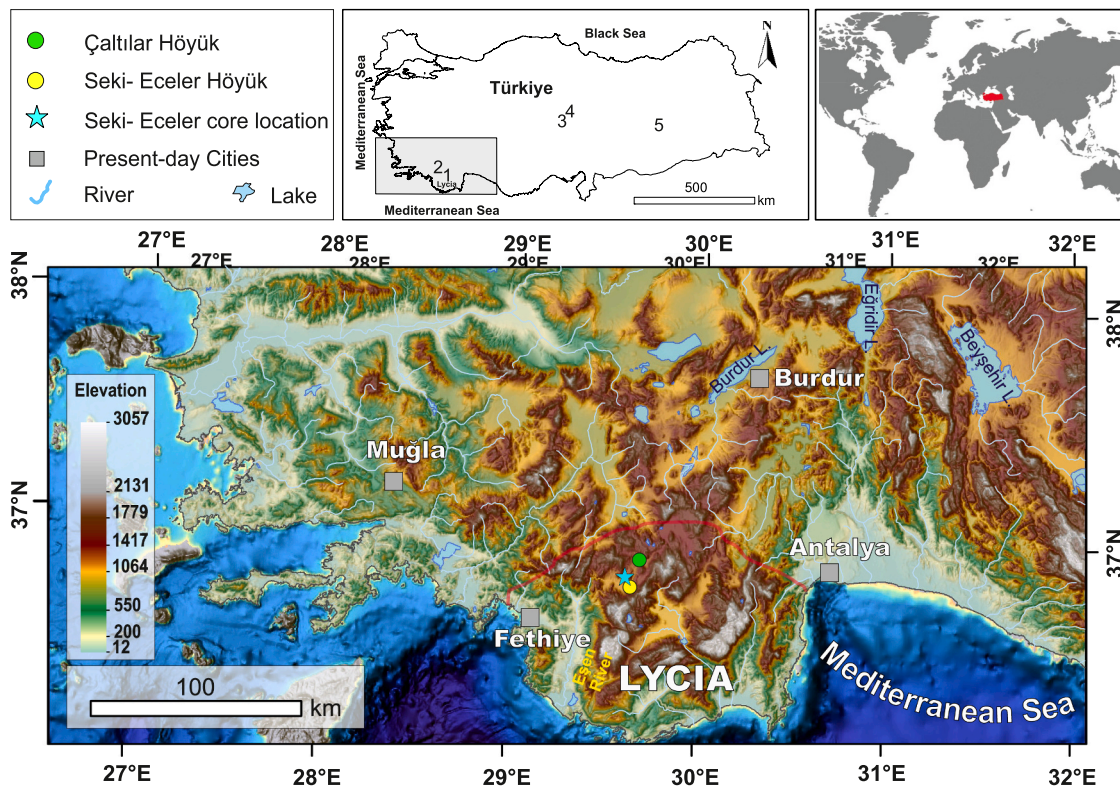


Fig. 1. Map showing the location of the study area and the regions whose proxies are compared in the correlation diagram: 1- Seki-Ecelers Plateau (this study), 2- Lake Gölhisar, 3-Lake Nar, 4-Lake Eski Acıgöl, and 5-Lake Hazar.

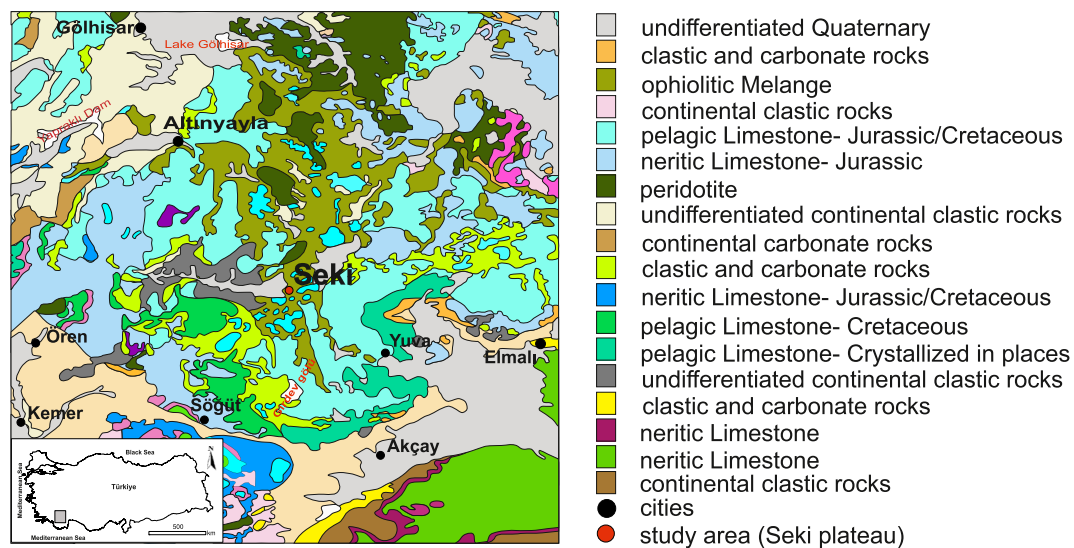


Fig. 2. Geological map of the study area (modified from the 1:500,000 Scale Geological Inventory Map Series of Türkiye prepared by MTA). <https://www.mta.gov.tr/en/sayfalar/maps/images/1-500/DENIZLI.jpg>

inputs could result in apparent age discrepancies towards older radiocarbon ages. However, no independent reservoir correction could be applied due to the absence of paired terrestrial macrofossil dates. Instead, stratigraphic consistency and agreement with regional palaeoclimate records were used to evaluate the reliability of the chronology. Because three dates (102 cm, 210 cm and 309 cm) were considered stratigraphically inconsistent, likely reflecting such contamination or reworking effects, they were down-weighted using a heavy-tailed Student-t likelihood, whereas the remaining dates were assigned near-normal likelihoods (Table 1). All radiocarbon determinations were

calibrated with IntCal20 (Reimer et al., 2020) with the Rbacon package (Blaauw and Christen, 2011) in R (R Core Team, 2024), and the model was additionally constrained by a core-top calendar-age tie point of -60 cal yr BP. Sedimentation priors were kept at default settings, except for memory, for which a relatively high prior mean was specified to reflect the continuous accumulation of fine lacustrine sediments. Posterior ages are reported as medians with 89% highest posterior density intervals. Ages below the deepest dated level were extrapolated using Bacon's prior accumulation structure (Fig. 3).

Table 1

AMS radiocarbon and calibrated ages of core JARK14–04. All measurements are on bulk sediment total organic carbon (TOC). Ages are reported as radiocarbon yr BP (relative to 1950 CE) $\pm 1\sigma$. The last column lists the per-date Student-t likelihood hyperparameters (t.a, t.b) used in rbacon to reflect differing reliability: (33, 34) approximates a near-normal likelihood, whereas (3, 4) is heavy-tailed to down-weight potential outliers. No per-sample calibrated ages are given here; calibration is performed only within the Bayesian age-depth model (see Methods). A calendar-age tie point at the core top (0 cm, –60 cal yr BP, $\sigma = 10$ yr) was applied separately and is not included in this table. TOC stands for Total Organic Carbon.

Lab code	Dated material	Core depth (cm)	^{14}C years BP	(t.a, t.b)
Poz-170065	TOC	75	3125 $\pm$ 35	(33, 34)
Poz-170066	TOC	102	4210 $\pm$ 30	(3, 4)
Poz-169999	TOC	197	4210 $\pm$ 30	(33, 34)
Poz-95796	TOC	210	4005 $\pm$ 35	(3, 4)
Poz-95794	TOC	250	6460 $\pm$ 40	(33, 34)
Poz-170067	TOC	309	2300 $\pm$ 30	(3, 4)
Poz-95768	TOC	340	7430 $\pm$ 50	(33, 34)
Poz-170228	TOC	377	9390 $\pm$ 60	(33, 34)
Poz-170095	TOC	418	9660 $\pm$ 60	(33, 34)
Poz-95767	TOC	450	10050 $\pm$ 70	(33, 34)

2.3. Itrax XRF (X-ray fluorescence) core scanner

The cores were analysed for element composition at a resolution of 0.5 mm via an Itrax M-XRF core scanner equipped with X-ray fluorescence (XRF)-EDS, X-ray radiography and an RGB color camera at the ITU-EMCOL core analysis laboratory (Croudace et al., 2006; Thomson et al., 2006; Çağatay et al., 2019). The X-ray generator was operated at 30 kV and 50 mA, with a counting time of 20 s. The element profiles were used as palaeoenvironmental proxies, including μ -XRF Sr, K, Ca, Ti, Fe, and Mn. Calcium (Ca) is associated with productivity and detrital input (Cohen, 2003). The amount of calcium in lake sediments is often used as an indicator of lacustrine productivity and changes in water levels (Croudace et al., 2006). It is commonly derived from endogenous calcite precipitation, which is enhanced during periods of increased

productivity and evaporation. This suggests warmer and/or drier climate conditions. Iron (Fe) and manganese (Mn) are frequently used to infer redox conditions at the sediment–water interface and therefore reflect a lake's oxygenation state (Thomson et al., 2006). Under anoxic conditions, manganese and, to a lesser extent, iron become soluble and are mobilised. This can lead to lower Mn and Fe concentrations in the sediment, whereas oxic conditions favour their precipitation, resulting in higher concentrations. Titanium (Ti) and potassium (K) are widely used as proxies for detrital input from catchment areas (Croudace et al., 2006; Thomson et al., 2006). High concentrations of Ti and K often indicate increased erosion and terrestrial runoff, which can be linked to increased precipitation, glacial melt or reduced vegetation cover in the surrounding area. Titanium (Ti) is an immobile element related to detrital input (Fischer and Wefer, 2012). Therefore, the Ca/Ti ratio normalises the carbonate content with respect to siliciclastic input (Croudace et al., 2006; Çağatay et al., 2019). The Sr/Ca ratio indicates the salinity and temperature of carbonates (Russell et al., 2004; Rosenthal et al., 2006). A higher Sr/Ca ratio indicates a greater abundance of aragonite than calcite and suggests dry climate conditions (Croudace et al., 2006).

2.4. TOC–TIC analysis

Total organic carbon (TOC) and total inorganic carbon (TIC) analyses were performed on the sediment core. This analysis provides important information about the palaeolimnology (organic production, evaporation/precipitation) of Seki-Eceler. The total organic carbon (TOC) and total inorganic carbon (TIC) contents were analysed via a Shimadzu TOC/TIC analyser at the ITU-EMCOL core analysis laboratory, with a sampling interval of at least 10 cm. The samples were freeze-dried, ground and weighed (48–52 mg). TOC was calculated as the difference between total carbon (TC) and TIC ($\text{TC} - \text{TIC} = \text{TOC}$). The TC analysis involved burning the samples at 900 °C to measure CO_2 , whereas the TIC analysis involved treating the samples with phosphoric acid at 200 °C to measure CO_2 . The TC calibration standard was potassium hydrogen phthalate, and the TIC calibration standard was sodium

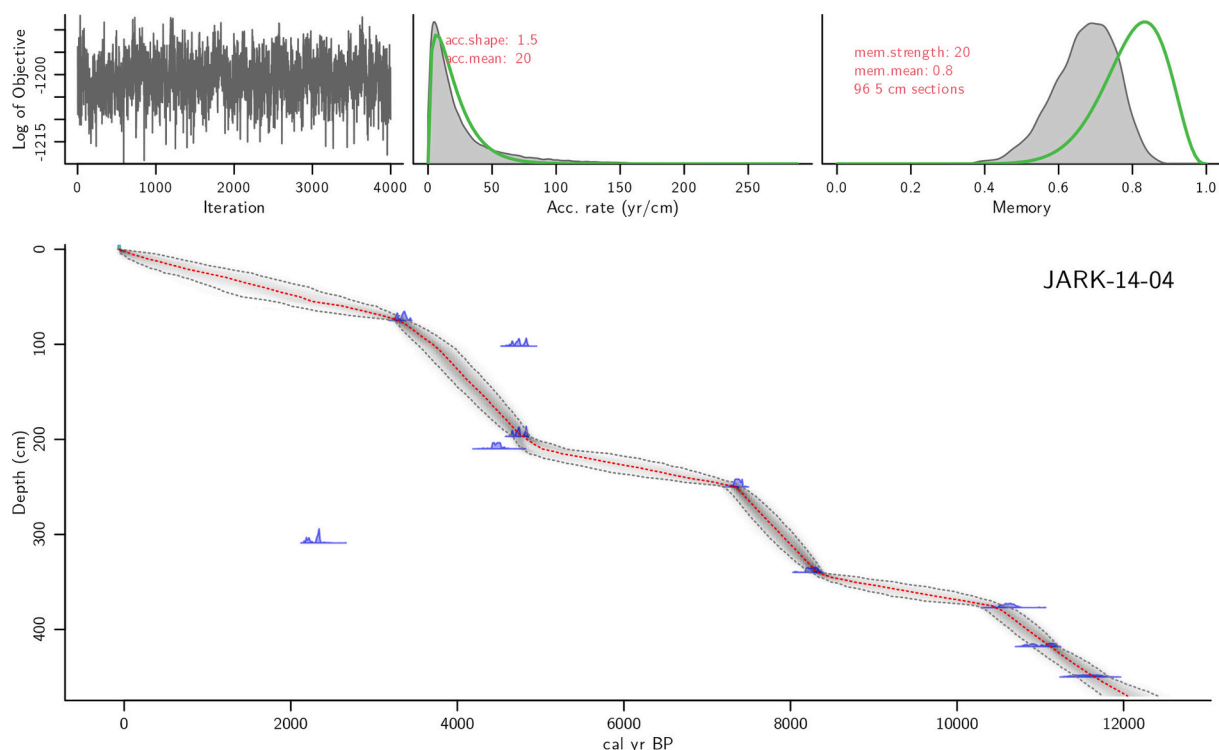


Fig. 3. Age–depth modelling of core JARK14–04 from Seki-Eceler.

hydrogen carbonate (NaHCO_3). The analyses have a precision better than 2% at the 95% confidence level.

2.5. Pollen analysis

For high-resolution pollen analysis, sediment samples were collected at 10 cm intervals and processed according to standard protocols (Cour, 1974). This included using cold HCl and HF to remove carbonates and silicates, respectively. Zinc chloride (ZnCl_2) was then used to separate palynomorphs (density > 2), after which the samples were sieved through 200 μm and 10 μm mesh. The pollen residues were mounted in glycerine and analysed via transmission light microscopy at 630 $\times$ and 1000 $\times$ magnification (with oil immersion). For each sample, a minimum of 300 terrestrial pollen grains were counted, with *Pinus* excluded from this total to prevent its high abundance from distorting the percentages of other taxa. This is a common practice in palynology (Heusser, 1988; Suc and Drivaliari, 1991). However, in most cases, even a minimum count of 67 pollen grains can reliably support palaeovegetation reconstructions based on lake sediment pollen percentages (Djamali and Cilleros, 2020). The percentages of all other pollen taxa were calculated on the basis of the sum of 300 pollen grains. We also counted *Pinus* separately and used the same total for its percentage calculation but did not include it in the AP (arboreal plants) curve. Taxa were identified via a pollen atlas (Reille, 1992; Beug, 2004) and unpublished photographs, with a minimum of 300 grains of terrestrial pollen counted per sample. Pollen data were visualised via TILIA (Grimm, 1987; Grimm, 1991–2015). A stratigraphically constrained cluster analysis using a square root transformation was run by CONISS (constrained incremental sum-of-squares cluster analysis). CONISS analysis was performed on the basis of changes in the percentages of Poaceae, Asteraceae Cichorioideae, and the AP and NAP values.

2.6. Pollen-plant richness and diversity

The level of detail in the pollen identification (family, genus or group of genera) in the established publications (Reille, 1992; Beug, 2004) enabled us to compare the pollen morphology. The Shannon–Wiener index is a widely used metric in ecology for measuring community diversity. It accounts for both richness (the number of different species) and evenness (the relative abundance of each species; Shannon, 1950). A higher value indicates greater diversity. Pielou's evenness index focuses specifically on the evenness of a community. It measures how equally abundant the species are. A value of 1 means that all species are equally abundant, whereas a value closer to 0 indicates that a few species dominate. The above indices, which are based on pollen percentages, were calculated via the “vegan” package in R (version 4.1.2) (Oksanen et al., 2018).

2.7. Statistical analysis (PCA)

Principal component analysis (PCA) is a statistical technique used to create composite proxy records by identifying the main factors driving the variability of the proxies. PCA was conducted on X-ray fluorescence (XRF), pollen and total organic and inorganic carbon (TOC–TIC) data via PAST software (Hammer, 2001). PCA works by creating new variables, known as principal components, which are linear combinations of the original variables. These principal components can be meaningfully interpreted and are designed to capture the greatest possible variance in the data. The first principal component accounts for the largest possible variance, the second accounts for the next largest variance, and so on. The analysis is based on a correlation matrix showing the linear relationships between the variables. This enables PCA to extract and highlight the most influential multivariate components that explain the variability within the original dataset. In essence, PCA transforms the data into a new set of uncorrelated variables, revealing underlying patterns and structures in the process. Due to the varying sampling

resolutions of the datasets (μXRF at 0.5 mm and pollen and TOC–TIC at 10 cm intervals), all proxy data were standardised to a common depth framework prior to PCA. To achieve this, the pollen and TOC–TIC datasets, which had the lowest resolution, were used to define the reference depth intervals. The μXRF data, which are of a higher resolution, were then resampled to these intervals by calculating the arithmetic mean of all μXRF measurements falling within each corresponding 10 cm depth bin. This binning approach reduces noise and ensures that the geochemical signal is directly comparable to the lower-resolution datasets. Upscaling (interpolation) of the pollen or TOC–TIC data was not applied to avoid introducing artificial variability. Instead, all datasets were aligned by downscaling the μXRF data to the coarser resolution. Prior to principal component analysis (PCA), all variables were log-transformed where necessary (e.g. elemental ratios) and standardised (z-score normalisation) to remove scale differences and ensure equal weighting among proxies. The resulting integrated dataset therefore consists of synchronised proxy values at consistent depth intervals, enabling robust multivariate comparison.

3. Results

3.1. Chronology and lithology

According to the age–depth model, the bottom of the core dates back to nearly 12000 yr BP and coincides with the Younger Dryas period (Fig. 3). Two bottom subsamples at depths of 450 and 418 cm were sampled from the brown clayey sand silt layer of the core (11600 yr BP and 11080 yr BP, respectively). The first part of this deposited layer is related to the transition from the Younger Dryas to the early Holocene (450 cm). A reddish-brown sandy clayey silt from 390 cm to 225 cm covers the previous layer and dates back from 10700 yr BP to 6100 yr BP. This layer was deposited during the early to the middle Holocene. In the middle of this layer, two other layers were recorded, consisting of grayish brown clayey silty fine sand and reddish clayey silty fine sand with spots. These layers were deposited from 8300 yr BP to 7950 yr BP. In addition, black sandy clayey silt with a root-like structure was observed on top of this silty layer, with an age of 5800 yr BP at the top of the layer (220 cm). This layer is overlain by a reddish dark brown sandy clayey silt with carbonated pebbles, which was deposited from 5800 yr BP to 4900 yr BP. Another thin layer of reddish brown clayey silty fine sand with 1–8 mm orange spot sediments overly the previous area, with an age of 4900 yr BP in the lower part and an age of 4800 yr BP in the upper part. There is an unconformity at 197 cm dated back to 4800 yr BP, which was sampled for age–depth modelling. From 197 cm to 130 cm, the core comprises dark brown clayey silty fine sand with black reduction spots, which are characterised by black chert or obsidian pebbles and some flat, gray ceramic layers (Fig. 4). The upper threshold of this layer has an age of 3950 yr BP. Another unconformity is recorded at this point. The uppermost part of the layer is a thick layer, approximately 130 cm, of dark brown, sandy clayey silt with rare subrounded pebbles. The uppermost subsample were selected at 75 cm depth, which coincides with the age of 3300 yr BP. The mean accumulation rate for this core is approximately 0.5 mm/yr or 20 yr/cm.

3.2. Geochemical analysis

3.2.1. Total organic carbon-total inorganic carbon (TOC–TIC) analysis

The TOC content exhibited relatively high fluctuations between 0.3 and 1.2 wt% along the core length (Fig. 5). After declining to a minimum of 0.3 wt% at 410 cm depth, the TOC content increased and peaked at 1.2 wt% at 340 cm depth, the maximum amount of total organic carbon (TOC) in the entire core. The maximum percentage of the TOC at 340 cm. This peak coincides with the lowest TIC value recorded in the core. Following this point, the TIC increases gradually to a maximum of 1.7 wt % at 130 cm depth and then decreases sharply to 0.5 wt% at 80 cm depth. The TIC peak coincides with a gradual decrease in the TOC

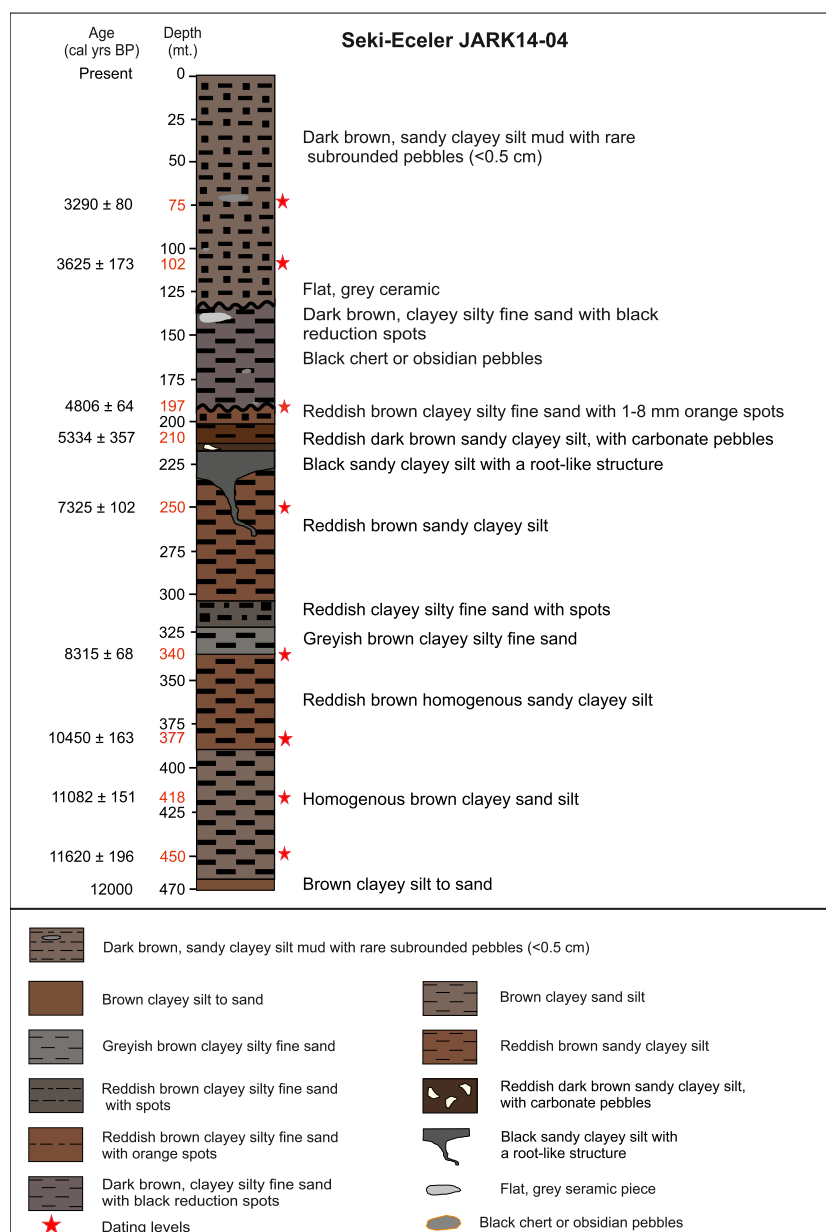


Fig. 4. Lithology of the core JARK14-04. Red stars indicate ^{14}C sampling levels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

content to 0.5 wt% at 80 cm. However, the TIC slightly increased from 0.5 wt% to 0.8 wt% at 80 cm depth. The ascending trajectory of TOC is inversely correlated with the decreasing trend of the TIC at depths between 80 and 0 cm (Fig. 5).

3.2.2. X-ray fluorescence (XRF) analysis

The μXRF scan of the Seki-Eceler JARK14-04 core revealed geochemical variations in the core during the past 12,000 yr BP (Fig. 5). The μXRF K, Ca, Ti, Mn, Fe, and Sr (cps) contents were measured. In addition to the elemental counts, the Ca/Ti, Mn/Fe, and Sr/Ca profiles were used as geochemical proxies. The μXRF K, Ca, and Sr profiles correlate well with each other. Ca and K fluctuate from 0 to 5200 cps between depths of 395 and 405 cm and considerably increase at 370 cm, and after some fluctuations, they significantly decrease at 350 cm. The μXRF profile of Ca increases again and reaches a peak of 5500 (counts per second -cps) at 320 cm, followed by K at 5500 cps. High-frequency fluctuations in μXRF Ca and K occur between 250 and 180 cm, with a maximum percentage of Ca occurring at 200 cm depth, which is the

highest percentage of the whole core, and sudden decreases in K occur at a depth of 190 cm. However, the μXRF K dramatically decreases at 80 cm. The μXRF Sr has a trajectory similar to that of these elements, with fewer fluctuations than those of the other elements (Fig. 5).

In contrast to μXRF Ca, K, Ti and Fe suddenly decrease between 380 and 350 cm and gradually decrease between 330 and 300 cm. However, an increase in Ti and Fe amounts to the maximum point at 300 cm was observed. The other two dramatic decreasing trends in μXRF Ti and Fe occurred at depths of 200 cm and 80 cm. The Ca/Ti and Mn/Fe ratios correlated positively with the μXRF Ca and K profiles. There is a positive correlation between the Sr/Ca, Ti, and Fe trajectories. (Fig. 5).

3.3. Palynological analysis

The 470 cm-long Seki-Eceler core (JARK14-04) encompasses a period from 12,000 yr BP to the present day (1950 CE). The pollen diagram consists mainly of herbaceous taxa, followed by coniferous trees, temperate broadleaf and needle-leaf trees, Mediterranean xerophytes

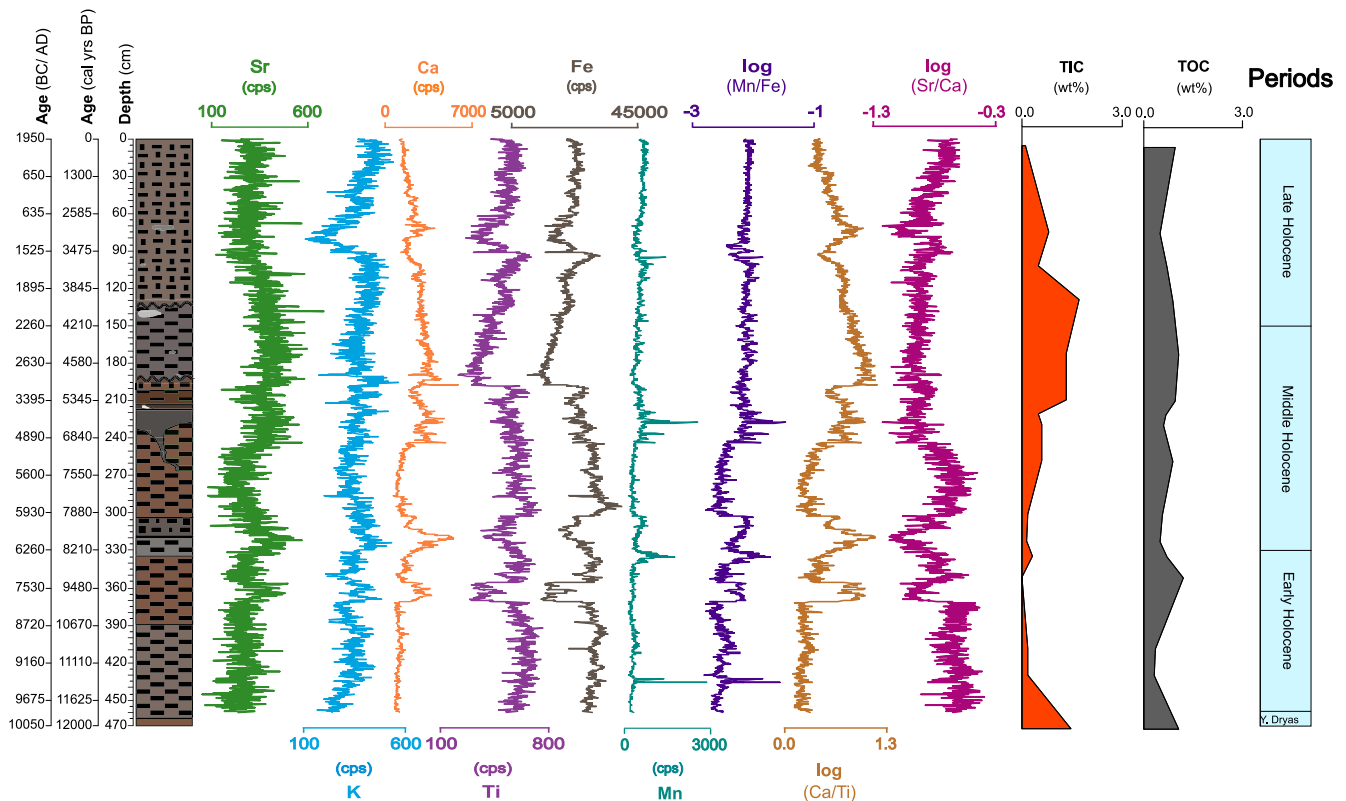


Fig. 5. μ -XRF geochemical element and element ratio profiles (K, Ca, Ti, Mn, Fe, Sr, Ca/Ti, Mn/Fe, and Sr/Ca) and total organic carbon–total inorganic carbon (TOC–TIC) profiles for core JARK14–04. Note that the element and element ratio profiles are in cps (counts per second) units. The μ XRF data were acquired at a high resolution of 0.5 mm, whereas the TOC–TIC measurements were obtained at a lower resolution of 10 cm sampling intervals. Therefore, the apparent differences in variability between the datasets reflect this disparity in sampling resolution to some extent.

and steppe herbs. The detailed pollen diagram comprises five pollen zones and six subzones. It includes 73 taxa, primarily herbaceous plants (Table 2, Figs. 7 and 8).

3.3.1. Pollen zone SE-1 (467–377 cm: 12000–10500 yr BP)

This zone is characterised by a high abundance of Asteraceae (Cichorioideae and Asteroideae), Poaceae, *Pinus* and *Artemisia*. Although *Artemisia* peaks at 28.3% at 433 cm, its highest value in the core shows a decreasing trend. The stable occurrence of *Juniperus* (0.4–3.1%) and deciduous *Quercus* (0–7%) indicates the presence of widespread coniferous and warm-temperate forests. The sporadic presence of taxa such as *Quercus ilex-coccifera*, *Pistacia*, *Ilex*, *Fraxinus*, *Castanea* and *Cedrus* indicates their limited distribution (Figs. 7 and 8).

3.3.2. Pollen zone SE-2 (377–257 cm: 10500–7400 yr BP)

This pollen zone is divided into two subzones: SE-2a: 377–345 cm; 10,500–8600 yr BP; and SE-2b: 345–257 cm; 8600–7400 yr BP.

3.3.2.1. Subzone SE-2a (377–345 cm: 10500–8600 yr BP). *Pinus* dominated this subzone, peaking at 60.9% at approximately 8600 yr BP. While herbaceous taxa such as Asteraceae Cichorioideae and Poaceae show negligible fluctuations, the pollen of Apiaceae suddenly increased to 31.3% at 357 cm. The total nonarboreal pollen (NAP) content remained consistently high (88.7–98.3%). Deciduous *Quercus*, Cupressaceae and *Juniperus* are present in low percentages (Figs. 7 and 8).

3.3.2.2. Subzone SE-2b (345–257 cm: 8600–7400 yr BP). The herbaceous taxa Asteraceae Cichorioideae, Poaceae and Asteraceae Asteroideae fluctuate dramatically. Moreover, *Artemisia* and *Centaurea solstitialis* gradually increased. Coniferous trees such as *Pinus*, *Juniperus*, *Cupressus* and *Abies* are prominent, but *Pinus* significantly decreases to

8.7% before increasing notably to 66.2%. Warm-temperate species, primarily deciduous oak (*Quercus*), decline and exhibit low fluctuations. The presence of Oleaceae, *Phillyrea* and *Quercus ilex-coccifera* indicates the presence of Mediterranean xerophytes. Steppe species such as *Ephedra* are present, but *Artemisia* is more prevalent (Figs. 7 and 8).

3.3.3. Pollen zone SE-3 (257–203 cm: 7400–5000 yr BP)

This zone is divided into two subzones: SE-3a: 257–237 cm; 7400–6600 yr BP; SE-3b: 237–203 cm; and 6600–5000 yr BP.

3.3.3.1. Subzone SE-3a (257–237 cm: 7400–6600 yr BP). This subzone is characterised by a decrease in Poaceae, Asteraceae Cichorioideae and Asteraceae Asteroideae. In contrast, the abundance of *Centaurea solstitialis* increased abruptly, reaching a core-wide peak of 33.8%. NAP taxa dominated until 6600 yr BP, after which the percentage of arboreal pollen (AP), particularly from *Pinus* and deciduous *Quercus*, increased rapidly (Figs. 7 and 8).

3.3.3.2. Subzone SE-3b (237–203 cm: 6600–5000 yr BP). The most abundant herbs are Poaceae, Asteraceae Cichorioideae and Asteraceae Asteroideae. The abundances of Poaceae and Asteraceae Asteroideae peak dramatically before they decline. *Pinus* and deciduous *Quercus* dominate the AP taxa, with *Pinus* accounting for 79% of the AP taxa. Steppe and hygrophyte species are present in low percentages (Figs. 7 and 8).

3.3.4. Pollen zone SE-4 (203–91 cm: 5000–3500 yr BP)

A key feature of this zone is the increase in *Centaurea solstitialis*, Asteraceae Asteroideae, and Amaranthaceae, which coincides with a significant decrease in Asteraceae Cichorioideae and Poaceae. The NAP taxa fluctuate, remaining consistently high. *Pinus* dominated the

Table 2

Summary of the relationship between human activity, arboricultural practices and pollen-based anthropogenic indicators in the Seki Basin from the Late Palaeolithic period to the present day. The table combines a chronological framework, archaeological phases and key pollen indicators (e.g. grazing and disturbance taxa, cultivated tree pollen).

Geologic Time	Time Period	Historical Periods	Vegetation Type	Climatic conditions	Climatic events	Human influence
Middle Holocene to Late Holocene	1750 yr BP to Present	Byzantine period/ Dark age, Selçuk/Beylik period, Ottoman period, Republic of Türkiye	oak-Coniferous forests/ Maquis	warm and dry	Little Ice Age Medieval climatic anomaly Dark Age cold period Roman warm period	Decrease in pastoralism Increase in Grazing No cultivated trees
	3500–1750 yr BP	Early Iron Age, Archaic period, Classical, Roman Imperial period	Xero Euxinian steppe-forest	warm and dry	Beyşehir occupation phase	Peak in pastoralism low Grazing Peak in Cultivated Trees
	5000–3500 yr BP	Bronze Age		cold and dry	cold and dry	Peak in pastoralism low Grazing No cultivated trees
Early to Middle Holocene	6600–5000 yr BP	Late Chalcolithic	Expansion of forests	warm and wet	Mid-Holocene Optimum Period	High fluctuations of grazing indicators, lower ratio of pastoral indicators No arboriculture
	7400–6600 yr BP	Early Chalcolithic	Expansion of forests and an increase in biodiversity			
	8600–7400 yr BP	Late Neolithic	Herbs are dominant, and AP value is scarce	cold and dry	8.2 cold event	Decrease in pastoral, grazing, and Stopping of the arboriculture
Younger Dryas to Early Holocene	10500–8600 yr BP	Late Mesolithic-Early Neolithic	open vegetation with scarce tree stands, and Marsh vegetation near the site	Oscillation of cold, dry, and warm, wet climatic conditions		Decrease in pastoral elements, grazing indicators, and cultivated trees.
	12000–10500 yr BP	Palaeolithic to early Mesolithic	Steppe with scarce trees to Open vegetation with scattered tree stands	Transition from an arid, cold climate to a warm, wet climate	Early Holocene climatic oscillations	decrease in pastoral, Peak in grazing, and Emergence of trees

arboreal component, peaking at 79.8% at the end of the zone despite an earlier decline. Deciduous *Quercus* and *Quercus ilex-coccifera* also exhibited minor fluctuations (Figs. 7 and 8).

3.3.5. Pollen zone SE-5 (91–1 cm: 3500- subrecent)

This zone is studied in two parts:

3.3.5.1. Subzone SE-5a (91–41 cm: 3500–1750 yr BP). This subzone is dominated by herbaceous taxa such as Poaceae and Asteraceae Asteroideae, with *Pinus* being the main arboreal taxon. After a peak, Poaceae and Asteraceae Asteroideae decreased, whereas Asteraceae Cichorioideae and *Centaurea solstitialis* increased. *Pinus* and deciduous *Quercus* gradually declined, although *Pinus* remained above 50.2%. The diversity of Mediterranean xerophytes was greater than that in previous periods (Figs. 7 and 8).

3.3.5.2. Subzone SE-5b (41–1 cm: 1750 yr BP to present). The most abundant plant families are Apiaceae, Pinaceae, Poaceae and Asteraceae Cichorioideae. The abundances of *Pinus* and Apiaceae increased dramatically and reached 79.9% and 88.6%, respectively. Deciduous *Quercus* is present in low percentages. *Betula* first appears at the top of the diagram, and *Quercus ilex-coccifera* is the only Mediterranean xerophyte present. The NAP percentage peaked at 99.4% at 13 cm (500 yr BP) (Figs. 7 and 8).

3.4. Predicting plant diversity from pollen richness and diversity indices

The calculations revealed significant variations in species diversity, and evenness at different depths (Fig. 7). This suggests that the ecological community has undergone substantial changes over the period represented by the sediment core. In the uppermost layer of the core, the Shannon index was relatively low at 0.9326, with a Pielou evenness index of 0.2934 (Supplementary S1). This indicates that a

community with low diversity and uneven distribution is likely dominated by a few species. This pattern continues to a depth of approximately 50 cm. A notable shift occurs at approximately 116 cm, where the Shannon index peaks at 1.9401 and Pielou's evenness index increases to 0.6589. This suggests a period of high species diversity and a more balanced community structure. This trend of relatively high diversity and evenness, with minor fluctuations, persists in the subsequent layers. At the bottom of the core, the values of the Shannon and Pielou indices were low, possibly due to a lack of preservation or an environment that could not support the habitat requirements of the species (Fig. 7).

3.5. Principal component analysis (PCA)

A dataset of X-ray fluorescence (XRF), pollen and total organic and inorganic carbon (TOC-TIC) data from sediment samples was analysed via principal component analysis (PCA). The analysis revealed significant changes in the Ca/Ti and Sr/Ca ratios, which were found to be negatively correlated. The PCA results revealed strong correlations between the elements Ca, Mn, Sr and Ti, particularly with respect to the Sr/Ca ratio. These findings suggest that these elements are linked to detrital input, i.e., the material deposited through the erosion of rocks (Kylander et al., 2011). The analysis also grouped these elements together, indicating that they behave similarly and collectively contribute to detrital input. However, the detrital indicators Ti and Fe exhibited a negative correlation (Fig. 6). The PCA also revealed that Fe was the most dominant signal. On the PCA plot, elements with strong positive loadings on Principal Component 2 (PC2), such as Ca, Mn, Sr and Ti, were separated from elements with negative loadings on Principal Component 1 (PC1), including Fe, total inorganic carbon (TIC) and arboreal/nonarboreal pollen (AP/NAP). Ca had the most pronounced positive loading on the PC1 axis. This could indicate subordinate source rock geology, or it could be related to biomineralisation or independent detrital input from bedrock, which would explain why it is separated from the other

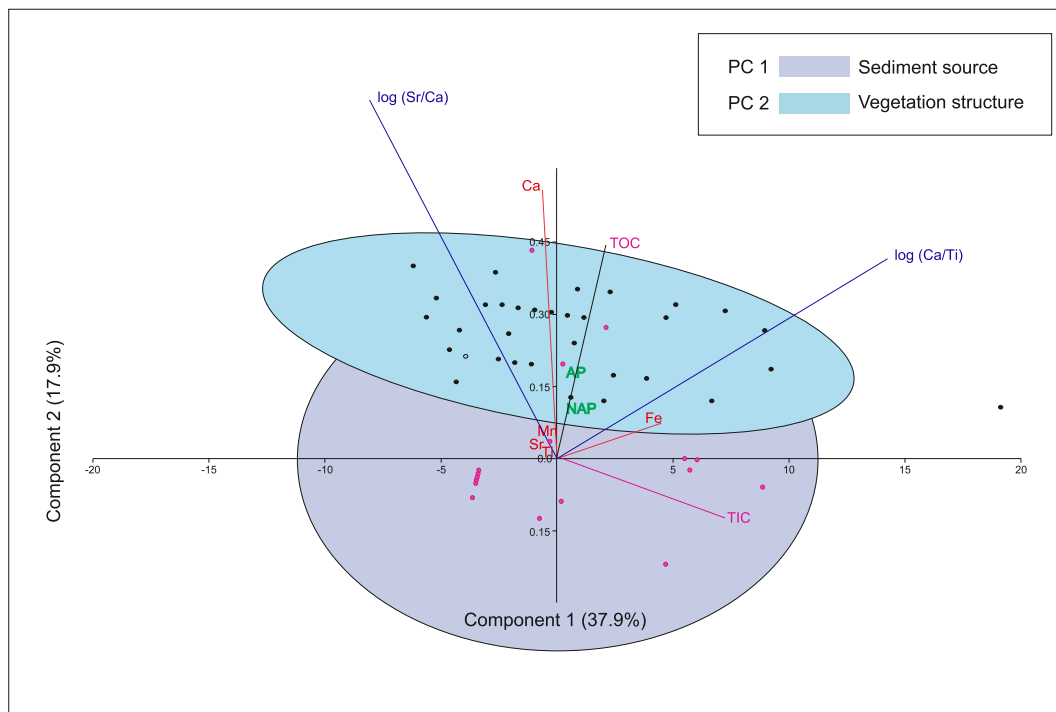


Fig. 6. Principal Component Analysis (PCA) biplot showing the relationships between pollen groups (AP and NAP), geochemical proxies (μ XRF Fe, Mn, Ca, Sr, Ti and $\log(\text{Ca/Ti})$ and $\log(\text{Sr/Ca})$) and organic matter variables (TOC and TIC). PC1 (37.9%) represents a gradient ranging from detrital input to carbonate productivity, and PC2 (17.9%) reflects the difference between open (non-arboreal) and forest (arboreal) areas. The arrows indicate the variable loadings and their lengths represent the strength of the correlation with the principal components. The shaded ellipse shows the overall distribution of samples in ordination space. High resolution μ XRF data (0.5 mm) were downsampled to 10 cm intervals using depth binning (mean values per interval), while pollen and TOC–TIC data were retained at their original resolution. No interpolation was applied.

elements.

4. Discussion

4.1. Palaeoenvironmental interpretations

This investigation uses pollen assemblages and geochemical data (μ XRF and TIC) to reconstruct the past environment of the Seki Basin, determining the interactions between vegetation, climate, and humans in the area from the Younger Dryas period to the present day (12000 yr BP to the present).

4.1.1. Late Younger Dryas (12000–11650 yr BP)

The palaeoenvironmental evidence from Seki-Eceler region suggests that the final phase of the Younger Dryas was not just a time of reduced tree cover; it was also a landscape where moisture limitations severely restricted both terrestrial biomass and catchment activity. The dominance of *Artemisia*, Asteraceae Cichorioideae, and *Centaurea solstitialis*, alongside very low arboreal pollen values, suggests the persistence of an open steppe environment under cold and dry conditions during this period. Additionally, the high representation of Poaceae indicates the widespread development of grass-dominated communities, which are typical of cold, moisture-limited steppe ecosystems in Anatolia during the Late Glacial (Bottema and Woldring, 2003; Roberts et al., 2011b). In such conditions, forest taxa were unable to expand beyond refugial or locally protected settings. In this context, the low AP values should be interpreted as evidence of suppressed woodland resilience rather than reduced pollen input alone. These findings are also supported by a high Ca/Ti ratio, a decrease in the Sr/Ca ratio, and lower levels of the elements μ -XRF K, Ti, Mn and μ -XRF Fe (Fig. 5). These results coincide with the lower $\delta^{18}\text{O}$ in Lakes Eski Acıgöl and Nar, and Soreq cave that is the sign of cold and dry weather in all over the region (Fig. 10).

4.1.2. Early Holocene (11650–8200 yr BP)

The early Holocene at Seki-Eceler was a period of significant transition, spanning Pollen Zones SE-1, SE-2a, and part of SE-2b. At Seki-Eceler, the early Holocene represents a fundamental reorganization of the local environment, marking the transition from a moisture-limited Late Glacial landscape to a more productive, hydrologically active Holocene ecosystem. The decline of *Artemisia* and the progressive expansion of deciduous *Quercus*, *Pinus* and *Juniperus* suggest that they indicate an increase in effective moisture, sufficient to support the establishment of woodlands, stabilisation of soils, and the gradual development of a more structured vegetation mosaic. Thus, the early Holocene can be considered a threshold period during which climatic improvement enabled the landscape to shift from open steppe dominance to mixed woodland and marsh environments (Figs. 7 and 8). This interpretation is strongly corroborated by the XRF data. High Sr/Ca ratios and elevated μ -XRF concentrations of Ti, K and Fe during this period indicate enhanced chemical weathering and increased detrital input (Fig. 5). This suggests a more active hydrological cycle with higher precipitation and runoff, which erodes soils and delivers terrestrial material to the lake. The high Sr/Ca ratio could also be attributed to the weathering of strontium-rich Mesozoic carbonate rocks in the drainage basin, as these rocks often have high strontium contents (Fig. 2). Furthermore, the presence of various aquatic herbs (Cyperaceae, *Nymphaea alba* and *Typha*) in the pollen record indicates the expansion of marsh vegetation, which also supports the idea of a wetter environment. The observed high-frequency oscillations between dry and humid phases in the Seki record are consistent with early Holocene variability documented in Anatolian speleothem archives, particularly from Kocain Cave, where rapid shifts in $\delta^{18}\text{O}$ values reflect short-term hydroclimatic instability (Jacobson et al., 2021), and also, from Dim Cave with a continuously positive water balance until 9000 yrs. BP (Ünal-İmer et al., 2016). This correspondence highlights the sensitivity of both lacustrine and speleothem systems to abrupt climate fluctuations in the eastern

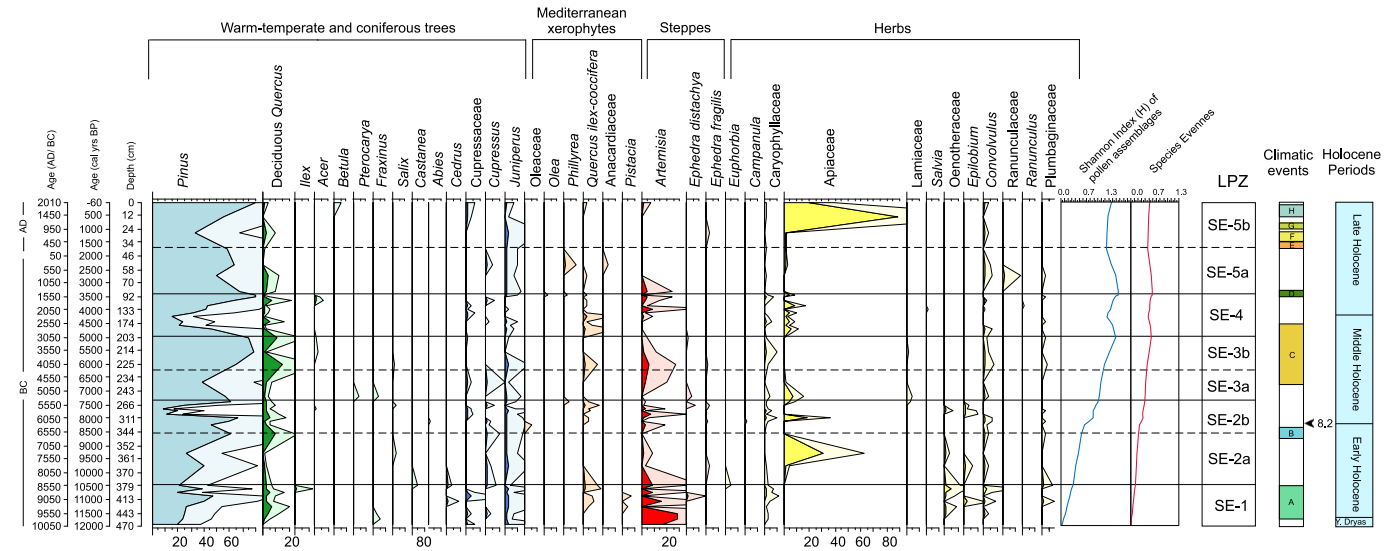


Fig. 7. The detailed pollen diagram shows the pollen percentages in the JARK14-04 sediment core (exaggeration, $\times 10$). A. Early Holocene climate change, B. Early Holocene Arboreal Maximum (EHPM), C. Middle Holocene Climatic Optimum, D. Beyşehir Occupation Phase (BOP), E. Roman Warm Period (RWP), F. Anatolian Dark Age Cold Periods (DACPs), G. Medieval Climatic Anomalies (MCAs), H. Little Ice Age (LIAs).

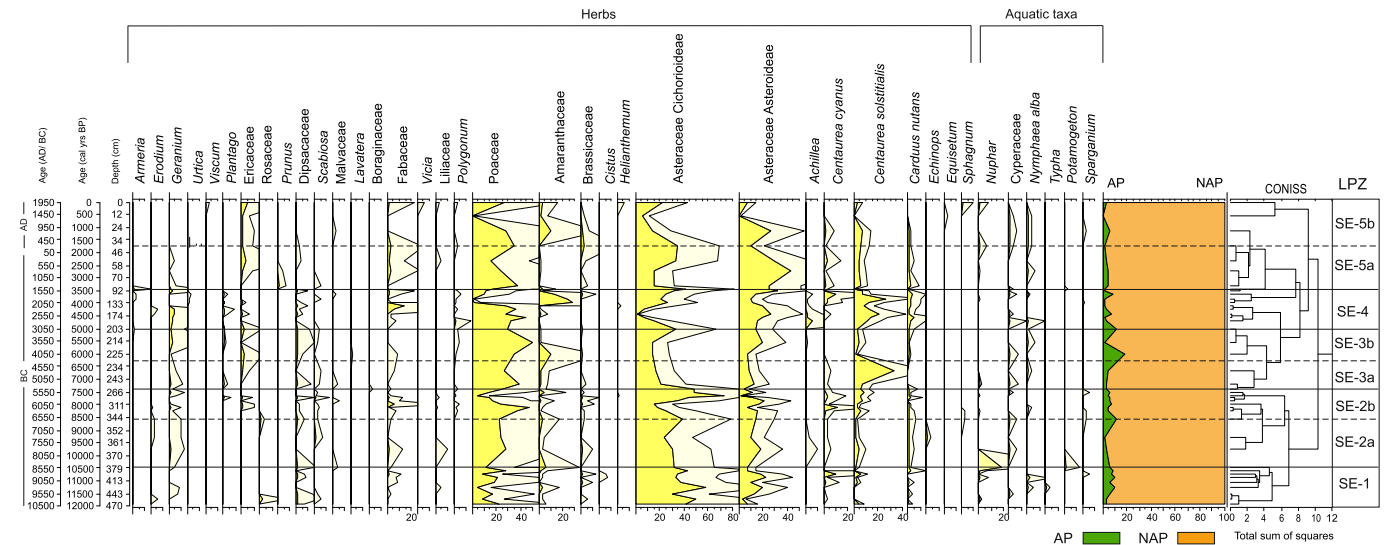


Fig. 8. The continuation of the detailed pollen diagram shows the pollen percentages in the JARK14-04 sediment core (exaggeration, $10\times$). The identification of pollen zones was carried out through the application of CONISS within the Tilia program, taking into account variations observed in Poaceae and Asteraceae Cichorioideae, as described by Grimm (1987).

Mediterranean. Between 11.6 and 10.5 yr BP, the alternation between steppe/grassland taxa and Mediterranean xerophytes/conifers indicates that the climate oscillated around an ecological threshold. At times, there was sufficient moisture availability to favour woodland expansion, whereas at others, the system reverted towards more open, drought-tolerant vegetation. The presence of *Pistacia* around 11300 yr BP is significant in this context as it suggests that the early Holocene warming period was characterised by increased summer dryness and milder winters, which is consistent with the region's emerging Mediterranean seasonality (Rossignol-Strick and Paterne, 1999). Rather than signalling full climatic stability, the presence of *Pistacia* indicates the establishment of a warmer, yet still seasonally water-stressed, environment. (Rossignol-Strick and Paterne, 1999). A subsequent shift between 10500 and 9300 yr BP revealed an increase in coniferous trees and high herbaceous plant diversity, indicating a transition from humid to colder conditions. While the pollen of herbaceous taxa such as Asteraceae Cichorioideae and Poaceae showed negligible fluctuations, that of

Apiaceae suddenly increased to 31.3% at 357 cm. This marked increase is particularly noteworthy as the Apiaceae family is commonly associated with open, disturbed and often dry habitats, including steppe and ruderal environments (Behre, 1981; Florenzano et al., 2014). This is supported by a decrease in both the AP and the Sr/Ca ratio, whereas the Ca/Ti ratio increases. The low concentrations of μ -XRF Ti, Mn, and Fe and μ -XRF K during this period, further, confirmed a reduction in detrital input, which was consistent with a period of decreased precipitation and reduced erosion. After 9300 yr BP, towards the end of the early Holocene, the proxy data indicate a clear shift in environmental conditions. The total organic carbon (TOC) values began to rise, suggesting an increase in biological productivity within the basin. This coincided with an increase in the Sr/Ca ratio and a corresponding decrease in the Ca/Ti ratio. Simultaneously, the increase in arboreal pollen (AP) values signals a notable expansion of forest cover in Lakes Hazar, Gölhisar, and Eski Acıgöl (Fig. 10).

4.1.3. Middle Holocene (8200–4200 yr BP)

This period consists of the SE-2, SE-3a, SE-3b, and SE-4 local pollen zones. The vegetation covers experienced three different successions. The first period (8200–6750 yr BP) was characterised by a low AP percentage, the second period (6750–4500 yr BP) was characterised by high AP values, and the third phase (4500–4200 yr BP) was characterised by a decreased AP content. The first of these phases (8200–6750 yr BP) reflects a period during which forest expansion was limited due to repeated disruption to woodland development caused by climate variability. The first period (8200–6750 yr BP) is characterised by three distinct phases of vegetation, with a clear multiproxy signal of the 8.2 ka event. This phase begins with a sharp decline in arboreal pollen (AP) between 8.3 and 8.2 ka, precisely coinciding with the global cooling event known as the 8.2 ka event (Wanner et al., 2011). The pollen record shows a surge in arid-indicator species, such as *Centaurea solstitialis*, *Apiaceae* and *Artemisia*, which confirms a shift to a cold and arid climate. Similar patterns have been reported in other Anatolian records, in which the presence of the *Apiaceae* family is interpreted as an indicator of steppe expansion (Bottema and Woldring, 2003; Roberts et al., 2011b). This interpretation is supported by geochemical data, which collectively point to a dry environment with reduced lacustrine productivity relative to detrital input and a decrease in TOC content (Figs. 5 and 10). In southwestern Türkiye, the environment was dominated by steppe forests and mountain coniferous forests at this time (van Zeist et al., 1975). Conversely, southwards shift of the ITCZ, associated with colder periods in the Northern Hemisphere (such as the 8.2 ka event), can lead to drier conditions in Anatolia. This is a common interpretation for the aridification signals observed in many Anatolian pollen and lake records during this event (Göktürk et al., 2011). In this context, the Seki record is consistent with a wider reorganization of atmospheric circulation in the eastern Mediterranean, characterised by reduced winter precipitation and increased seasonality, which limited both catchment runoff and biomass production. Following the initial arid phase, the Mid-Holocene Climatic Optimum occurred in this basin 6000 yr BP. This was characterised by a significant increase in both AP and overall plant biodiversity, indicating a major improvement in the climate (Fig. 10). The widespread development of warm-temperate, coniferous and Mediterranean xerophyte taxa suggests prolonged favourable conditions, which are likely warmer and wetter than those in the preceding phase. This sustained climatic stability enabled the expansion of forest cover across the landscape. This is also evidenced by a high Sr/Ca ratio and a decrease in the Ca/Ti ratio. The TOC content also increased during the Mid-Holocene Climatic Optimum. Taken together, these proxies suggest that the Mid-Holocene Climatic Optimum was a period characterised by enhanced hydrological efficiency and ecological stability. During this time, increased effective moisture promoted woodland closure, organic productivity, and greater landscape heterogeneity. This interval represents a sustained period in which the climate became sufficiently favourable to support long-term forest persistence, rather than merely reflecting warmer conditions. The third period (4500–4200 yr BP) marked a distinct return to arid conditions. Pollen data revealed a sharp decline in AP, particularly for *Pinus*, and a corresponding increase in steppe herbs and *Artemisia*. This shift suggests a change from forest cover to drier meadow vegetation with only scattered Mediterranean xerophyte trees, such as *Quercus ilex*. The collective evidence points to a rapid transition towards a drier and less favourable climate. While previous interpretations have often relied on records from the eastern Mediterranean, such as those from Soreq Cave, a more regionally relevant comparison is provided by Anatolian speleothem sequences. For instance, the $\delta^{18}\text{O}$ records from Kocain Cave clearly capture the 8.2 ka event as a sharp shift towards drier conditions, closely matching the timing and magnitude observed in the Seki multiproxy dataset (Jacobson et al., 2021). This strengthens the regional coherence of the signal and reduces reliance on archives in distant geographical locations. Beyond the 8.2 ka event, the variability observed in the Seki Basin during the mid-Holocene is consistent with the multi-centennial

oscillations recorded in Anatolian speleothems. For instance, reports from Dim Cave $\delta^{18}\text{O}$ records showed a rapid transition from wetter to drier conditions during 4320–3870 yr BP (Baykara et al., 2026). This suggests that the hydroclimatic instability observed during this period was not only regional, but also occurred at a high temporal resolution.

4.1.4. Late Holocene (4200 yr BP to present)

This period is characterised by a complex interplay of climatic fluctuations and increasing human impact, spanning the top of pollen Zone SE-4 and all of Zone SE-5: 4200–3500 yr BP, 3500–3300 yr BP (Beyşehir Occupation Phase) (Eastwood et al., 1998), 3300–1750 yr BP, 1750–1500 yr BP (Roman Warm Period, 200–400 CE) (Chen et al., 2011; Bakker et al., 2013; Ören, 2023), 1500–1150 yr BP (Dark Age Cold Period, 450–800 CE) (Helama et al., 2017; Jacobson et al., 2021, 2022), 1150–550 yr BP (Medieval Climatic Anomaly, 950–1400 CE) (Danladi et al., 2023), 550–110 yr BP (Little Ice Age, 1400–1800 CE) (Danladi et al., 2023), and 110–0 yr BP (1800–1950 CE). Unlike in the earlier Holocene, environmental change during this period cannot be explained by climate alone because human land use increasingly altered the vegetation pattern and sometimes overrode climatic trends. A low Ca/Ti ratio from 4200 to 3500 yr BP suggests more humid conditions which is consistent with Dim-3 stalagmite $\delta^{13}\text{C}$ results that showed the southwestern Türkiye was more humid after “4.2 megadrought event” (Baykara et al., 2026). This was followed by a period from 3500 to 3300 yr BP, which correlates with the Beyşehir occupation phase (BOP) (Eastwood et al., 1998). Although classic BOP indicator species are the high abundance of *Artemisia* and other anthropogenic indicators suggests increased human interference in the form of deforestation. This interval is important because of the difference between the climatic and vegetation proxies. Despite evidence of relatively favourable moisture conditions, the pollen signal shows that the landscape was opening up. This implies that human disturbance rather than deterioration in the climate was the main reason for the change in vegetation. The coastal regions of southwestern Anatolia such as Dim cave surrounded area were more humid than Seki region during this time (Baykara et al., 2026).

A significant shift occurred between 2750 and 1750 yr BP. A decline in AP, particularly *Pinus*, coincided with an increasing trend in Asteraceae families (Asteroideae and Cichorioideae). During this period, the high abundance of *Apiaceae* and the prominent presence of Poaceae suggest a landscape that is predominantly open and dominated by herbaceous vegetation. In contrast, arboreal vegetation was more dominant in some areas such as Taşeli Plateau in southern Anatolia (Karlıoğlu et al., 2016). In particular, *Apiaceae* is widely recognised as a sensitive indicator of disturbed soils, grazing pressure and anthropogenic landscape opening (Behre, 1981; Florenzano et al., 2014). The high number of Asteraceae Cichorioideae, which are known to be a marker for pastoralism and grazing (Florenzano et al., 2014), alongside the decreasing AP, suggests that large-scale deforestation was driven by humans rather than by natural climate change. The high percentages of *Apiaceae* and Poaceae further support the interpretation of intensified grazing and land use, as these taxa are commonly found in pastureland and in soils that have been trampled or enriched with nutrients in Mediterranean environments (Mercuri et al., 2013; Florenzano et al., 2014). This is an example of divergent evidence, a low AP signal that suggests anthropogenic activity rather than a colder or drier climate in this context.

From 1750 to 1500 yr BP, the climate became slightly warmer and drier, aligning with the Roman warm period (Chen et al., 2011). This is supported by a gradual increase in the AP ratio, including deciduous *Quercus*, and a relatively low Ca/Ti ratio. This interpretation is further supported by dendrochronological records from Anatolia, which provide evidence of annual hydroclimatic variability during the Late Holocene. Tree-ring-based reconstructions suggest relatively warm and, in certain regions, moisture-variable conditions during the Roman Warm Period. This was followed by a transition to cooler and more unstable climatic conditions during the Dark Ages Cold Period (Köse et al., 2011;

Touchan et al., 2005; Griggs et al., 2007; Baykara et al., 2026). Similarly, the Dark Ages Cold Period (1500–1150 years BP) shows a contrasting pattern in our records. Tree-ring chronologies from Anatolia and the eastern Mediterranean consistently document reduced growth rates during this period, reflecting cooler temperatures and/or decreased precipitation. This is consistent with broader regional climatic deterioration as inferred from speleothem records (Helama et al., 2017; Köse et al., 2011; Baykara et al., 2026).

The late Holocene was characterised by generally high and fluctuating values of TIC. TOC values are consistently low throughout the late Holocene. This suggests that, despite ongoing human activity, natural climatic signals of warming can still be registered in proxy records.

Crucially, the high-frequency climatic fluctuations observed in the Seki-Eceler record during the late Holocene closely align with speleothem-based reconstructions from Anatolia. Notably, the Kocain Cave record offers well-dated evidence of hydroclimatic variability on a centennial scale during pivotal periods such as the Roman Warm Period and the Dark Ages Cold Period (Jacobson et al., 2021). Similarly, studies on speleothems from Dim Cave shows a detailed climatic condition changes from the early Holocene (Ünal-İmer et al., 2016) and the Late

Holocene (Baykara et al., 2026). These oscillations are reflected in our pollen and geochemical data as shifts in arboreal pollen percentages and detrital input, suggesting a strong coupling between regional climate forcing and local ecosystem responses. This agreement with local, high-resolution speleothem archives is significant, as it shows that the variability observed in the Seki-Eceler record is part of a broader Anatolian climate pattern, not just a local signal. However, relying solely on distal records, such as those from Soreq Cave, can obscure regional heterogeneity, particularly during the late Holocene, when climatic gradients across the eastern Mediterranean became more pronounced.

4.2. Anthropogenic influences on the environment

Reconstructing past human activities from fossil pollen records is an essential yet challenging task in palaeoecology. This process is complicated by the difficulty of distinguishing between signals caused by human actions and those resulting from natural climate shifts. Deza-Araujo et al. (2020) investigated the effectiveness of various anthropogenic indicators in different regions, highlighting the importance of validating these proxies. Woodbridge et al. (2019) used pollen,

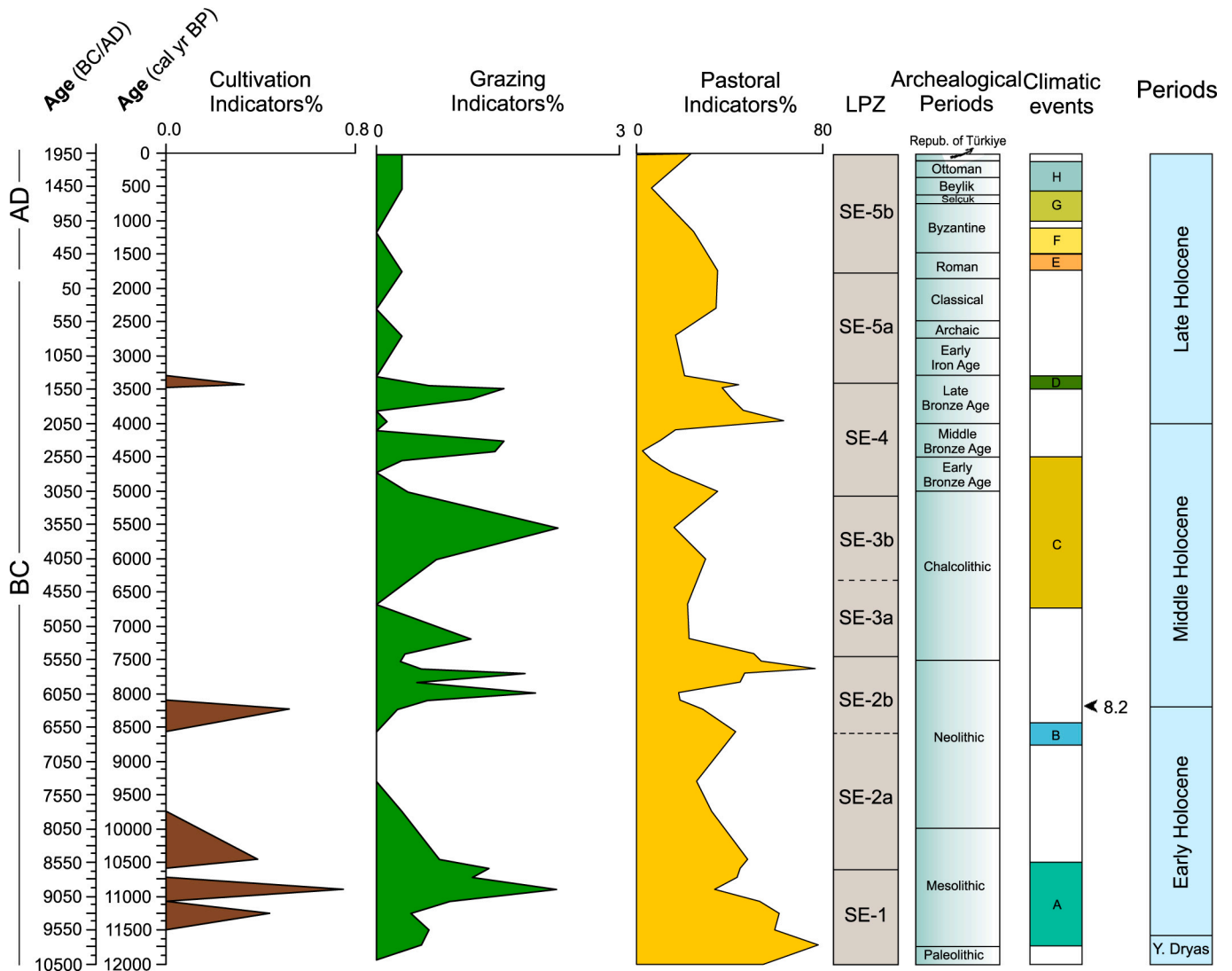


Fig. 9. Comparison of cultivated and anthropogenic plants to historical periods in Seki-Eceler. The Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago* were used as pastoral taxa; Caryophyllaceae and *Plantago* were used as grazing indicators; and *Castanea*, *Olea*, and *Pistacia* were cultivated species to conform to the anthropogenic chart. The LPZ is the local pollen zone. The letters are A: Early Holocene Climatic Oscillation, B: Early Holocene Arboreal Maximum (EHPM), C: Middle Holocene Climatic Optimum, D: Beyşehir Occupation Phase (BOP), E: Roman Warm Period (RWP), F: Anatolian Dark Ages Cold Period (DACP), G: Medieval Climatic Anomaly (MCA), H: Little Ice Age (LIA).

palaeoclimate and archaeological evidence to investigate the relationships between demography and vegetation change, and to explore how the development of agriculture varied spatially. When constructing our anthropogenic chart (Fig. 9), we categorised Asteraceae Cichorioideae, Amaranthaceae, Ranunculaceae, *Artemisia*, and *Plantago* as pastoral taxa; Caryophyllaceae and *Plantago* as grazing indicators; and *Castanea*, *Olea*, and *Pistacia* as cultivated species on the basis of Behre (1981) and Deza-Araujo et al. (2020).

4.2.1. Late Palaeolithic and Mesolithic Periods (11950–10000 yr BP; 10000–8050 BCE)

Examining anthropogenic indicator taxa reveals that human influence on the environment of the Seki Basin began during the late Palaeolithic period (~12000 yr BP), which coincided with the first evidence of a human presence at Sagalassos (Bakker et al., 2012). A key finding is that, although food plant cultivation began in the Near East and Levant by 10000 yr BP (van Zeist et al., 1975; Roberts et al., 2011a, 2011b, 2019; Asouti and Kabukcu, 2014), pollen data from the Seki region indicate that hunter-gatherer practices persisted (Turner et al., 2008; Fig. 9).

The pollen record shows a low but continuous presence of taxa such as *Pistacia*, *Olea* and *Castanea* during the transition from the late Palaeolithic to the Mesolithic; however, this should not be taken as proof of cultivation or arboriculture. Instead, they most likely reflect the natural regional vegetation and/or long-distance pollen transport signals common to these taxa in Mediterranean settings. Consistent with this interpretation, the early Holocene–Mesolithic interval is characterised by the absence of clear arboricultural activity (Table 2), and these taxa are therefore interpreted as components of the natural vegetation rather than indicators of human cultivation. Arboriculture and systematic land management practices are more plausibly associated with the socio-economic developments of the later Neolithic and Bronze Ages in Anatolia. The abundance of *Pistacia* in our record may be related to its use as firewood at nearby sites such as Çatalhöyük (Asouti and Kabukcu, 2014). Alternatively, its presence may simply reflect its ecological distribution in the region during favourable climatic conditions (Woodbridge et al., 2019).

4.2.2. Neolithic Period (10000–7500 yr BP; 8050–5550 BCE)

As early as 9750–8500 BP, humans began to impact the region's environment through pastoralism. However, during the period from 8500 to 8200 yr BP (6550–6250 BCE), cultivated and grazing species were introduced (Fig. 9). These changes coincide with the increase in population in Anatolia (Woodbridge et al., 2019), and adoption of agriculture in the eastern Mediterranean during the Neolithic period (Turner et al., 2008; Berger et al., 2016), as well as the emergence of the earliest agro-pastoral economies in the Old World (Asouti and Kabukcu, 2014). However, fruiticulture in the region declined from the late Neolithic period (8150 yr BP; 6200 BCE), whereas grazing and pastoral activities continued throughout the pollen diagram, albeit with several fluctuations (Fig. 8). Roberts et al. (2019) reported that the population of southern Anatolia peaked before 8000 yr BP (6050 BCE), but the 8.2 ka drought event seemed to have brought about a change in taxa, shifting from those typical of arboriculture to those more common in grazing and pastoralism. Similar changes in agricultural strategies caused by 8.2 ka droughts have been recorded across the eastern Mediterranean (Cruz-Silva et al., 2023). Forests were more dominant in southwestern Anatolia compared to oak parklands of central Anatolia merged by human (Asouti and Kabukcu, 2014; Woodbridge et al., 2019).

4.2.3. Chalcolithic Period (7500–5000 yr BP; 5550–3050 BCE)

During this time, the agricultural production of communities in the Fertile Crescent increased from the Chalcolithic period onwards (Roberts et al., 2011a). Our data show that grazing indicators fluctuated during the middle Holocene but levelled out in the late Holocene.

Furthermore, grazing and pastoral indicators suggest that animal husbandry became more prevalent than arboriculture by the Chalcolithic and Bronze Age (7000–3250 yr BP; 5050–1300 BCE), coinciding with the middle Holocene Climatic Optimum (see Fig. 8). Moreover, the number of settlements in the Sagalassos region increased during this period (Bakker et al., 2012). During this time, the distinctive regional human effect on the vegetation of southwestern Anatolia was recorded in Ağlasun (Woodbridge et al., 2019).

4.2.4. Bronze Age (5000–3250 yr BP; 3050–1300 BCE)

From 5000 yr BP onwards, a clear increase in human settlement and food production led to widespread deforestation (Yaalon, 1997; England et al., 2008; Bakker et al., 2012; Woodbridge et al., 2019). During this period, urban centres and empires such as the Akkadian arose, and some scholars have linked the collapse of the latter to the debated 4.2 ka megadrought (Cookson et al., 2019; Ön et al., 2023). The Middle Bronze Age, a period of significant urbanisation and trade in Anatolia, exhibits a distinct pattern in our region. The 13th century BCE is also the time of the flourishing of the ancient kingdoms and empires in the Mediterranean and Egypt (Kaniewski et al., 2015). The Beyşehir Occupation Phase (BOP) was the most significant period of human–environment interaction in the late Holocene of southwestern Anatolia (Kaniewski et al., 2007a). Although the complete palynological OJCV index (*Olea*, *Juglans*, *Castanea*, *Vitis*) is employed elsewhere, our record differs markedly from this classical pattern. *Juglans* is not present at all throughout the sequence, and *Castanea* only occurs sporadically and in very low percentages in the lower levels of the core (Fig. 7). Therefore, *Olea* is the only species that can be considered a potential indicator of arboricultural activity in the Seki region, despite being found in low quantities. The deviation of the Seki record from the classical BOP signal cannot be attributed to altitude alone. Several additional factors likely contributed to this pattern. The Seki-Eceler region is characterised by calcareous, clay-rich, marl-derived soils, which are often found alongside alluvial and colluvial deposits (Ryan et al., 2024). These conditions result in a relatively high soil pH and reduced drainage capacity, both of which are suboptimal for several OJCV taxa. *Juglans* requires deep soil. A high calcium carbonate (CaCO₃) content and an alkaline pH (>7.5) reduce the availability of essential micronutrients such as iron (Fe) and zinc (Zn). Elevated bicarbonate concentrations can also inhibit iron uptake by the roots, leading to chlorosis and reduced growth (Mengel, 1994; Begg et al., 1997). *Castanea* is highly sensitive to calcareous conditions and typically requires acidic, well-drained soils (pH <6.5). Numerous studies have demonstrated that chestnut is largely excluded from carbonate-rich substrates due to impaired nutrient uptake and reduced root development (Conedera et al., 2004). Therefore, its sporadic occurrence in the pollen record is more likely to reflect limited local cultivation or extra-local input than sustained agroforestry practices. Therefore, its sporadic occurrence in the pollen record is more likely to reflect limited local cultivation or external input than sustained agroforestry practices.

The absence of *Juglans* and the discontinuous presence of *Castanea* in the Seki-Eceler pollen assemblage are best explained by edaphic constraints rather than by cultural or altitudinal factors alone. This interpretation supports the idea that the Beyşehir Occupation Phase was expressed differently across the region, with local soil conditions playing a decisive role in shaping agricultural choices. This is in comparison with southern Anatolia that cultivation of *Juglans* was the result of the distinctive human activity in some areas such as Taşeli Plateau (Karlıoğlu et al., 2016).

However, there are some written evidences of famine and food shortage in the Mesopotamia and Anatolia during the late phases of Bronze Age (Kaniewski et al., 2015). Here, evidence of settlement is scarce, accompanied by a decline in pastoralism, grazing and cultivation (Momigliano et al., 2011a; Fig. 9). The reasons for this decline and any potential climatic role remain unclear. This time period called the “Late Bronze Age Collapse” (Kaniewski et al., 2015; Baykara et al., 2026).

4.2.5. Early Iron Age, Archaic and Classical (3200–1250 yr BP; 1250 BCE–700 CE)

Between 3250 and 2250 yr BP, open grasslands decreased sharply due to a combination of climatic and sociopolitical factors (Fig. 9). During the Hellenistic, Roman, and early Byzantine periods (2250–1250 yr BP), pastoralism and cultivation increased, with olive cultivation expanding across Anatolia (Vermoere et al., 2000). The number of settlements in Lycia declined sharply between 550 CE and 650 due to droughts, the Plague of Justinian, earthquakes, and warfare (Bakker et al., 2012; Dönmez et al., 2021). This period, often called the Anatolian Dark Ages, saw land abandonment and reduced arboriculture, which may have led to the increase in open vegetation (Kaniewski et al., 2007b; Bakker et al., 2012, 2013; Woodbridge et al., 2019; Jacobson et al., 2022). Notably, our data from Seki-Eceler does not show evidence of aridification during this time (Fig. 9).

4.3. Regional comparison of the environment

The Younger Dryas period in the Seki Basin is a prime example of regional climatic coherence. Our findings of a cold and arid climate dominated by steppe and grassland vegetation are not unique (Fig. 10). This pattern is consistent with that observed in other cores from southwestern Türkiye, such as those from the Söğüt and Karamık marshes (van Zeist et al., 1975), and reflects the presence of oak, pine and juniper forests as documented at the Öküzini Cave (Emery-Barbier and Thiébaud, 2005). More broadly, these similarities reflect the widespread impact of the Younger Dryas event across Anatolia, where grass predominated in Lake Eski Acıgöl (Woldring and Bottema, 2002), and where lake levels fell dramatically in both Lake Hazar (Ön et al., 2018) and Lake Van (Çağatay et al., 2014), due to increased evaporation. However, the apparent regional coherence of the Younger Dryas

signal must be critically evaluated in light of the methodological differences between the various archives. The Seki-Eceler chronology relies on radiocarbon dating of bulk sediment, which is particularly susceptible to hard-water reservoir effects, the incorporation of detrital or reworked carbon, and potential mixing within the sediment column. These processes can result in apparent ages that are systematically older than the true depositional age, particularly in carbonate-rich lacustrine systems. In contrast, speleothem records (e.g. Soreq Cave) are based on U–Th dating, which provides absolute chronologies with uncertainties typically on the order of decades to centuries. Consequently, chronological discrepancies of several hundred years between lake and speleothem records are anticipated, and apparent synchrony should be interpreted as indicating shared large-scale climatic influences rather than precise temporal alignment. The comparability of the Seki-Eceler record with multi-proxy data from Lake Eski Acıgöl, Soreq Cave and Lake Nar further demonstrates that our findings capture the dominant regional climatic signals of this period (Fig. 10). However, these comparisons should be interpreted with caution because the Seki-Eceler chronology is based on radiocarbon dates obtained from bulk sediment, whereas several of the speleothem records shown in Fig. 10 rely on high-precision uranium-thorium (U–Th) chronologies. Consequently, short chronological discrepancies may exist between the archives, particularly in lacustrine settings, where the incorporation of reworked or old carbon may influence bulk TOC dates. Therefore, the correlations presented here emphasise broad temporal agreement and shared climatic trends rather than exact synchronisation of individual events.

Additionally, the early Holocene transition in the Seki Basin reflects a widespread regional shift towards warmer, more humid conditions. The overall trend of increasing arboreal pollen (AP) and the expansion of coniferous and Mediterranean xerophyte forests is consistent with

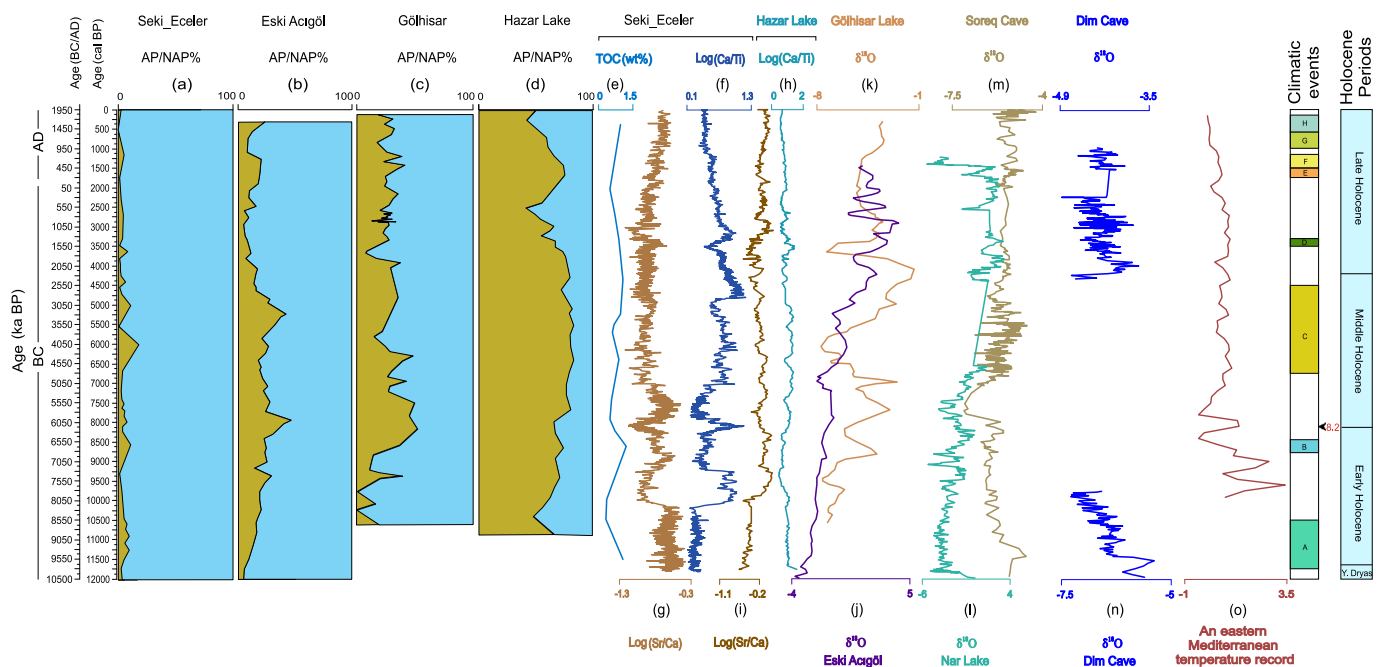


Fig. 10. Comparison of palynological record, μ XRF, and stable isotope data: a) pollen data of Seki-Eceler core in southwestern Anatolia (this study), b) pollen data of Lake Eski Acıgöl core in central Anatolia (Roberts et al., 2001), c) pollen data of Lake Gölhisar core in southwestern Anatolia (Eastwood et al., 1999), d) pollen data of Lake Hazar in eastern Anatolia (Biltekin et al., 2018), e) TOC profile of Seki-Eceler core in southwestern Anatolia (this study), f) log Ca/Ti profile of Seki-Eceler core in southwestern Anatolia (this study), g) log Sr/Ca profile of Lake Hazar in Eastern Anatolia (Ön et al., 2018), h) log Ca/Ti profile of Lake Hazar in Eastern Anatolia (Ön et al., 2018), i) log Sr/Ca profile of Lake Hazar in Eastern Anatolia (Ön et al., 2018), j) $\delta^{18}\text{O}$ profile of Lake Eski Acıgöl core in Central Anatolia (Roberts et al., 2001), k) $\delta^{18}\text{O}$ profile of Lake Gölhisar in southwestern Anatolia (Eastwood et al., 2007), l) $\delta^{18}\text{O}$ profile of Lake Nar in central Anatolia (England et al., 2008), m) $\delta^{18}\text{O}$ profile of Soreq Cave in the Levant (Bar-Matthews et al., 1999), n) $\delta^{18}\text{O}$ profile of Dim Cave in southwestern Anatolia during the early Holocene (Ünal-İmer et al., 2016) and the middle-Late Holocene (Baykara et al., 2026). o) The reconstruction of an eastern Mediterranean temperature record for the last 10000 yr BP. The letters are A: Early Holocene Climatic Oscillation, B: Early Holocene Arboreal Maximum (EHAM), C: Middle Holocene Climatic Optimum, D: Beyşehir Occupation Phase (BOP), E: Roman Warm Period (RWP), F: Anatolian Dark Ages Cold Period (DACP), G: Medieval Climatic Anomaly (MCA), H: Little Ice Age (LIA).

patterns observed in continental Anatolia, such as at Lake Eski Acıgöl (Roberts et al., 2001) and Lake Hazar (Biltekin et al., 2018). This correspondence, supported by high $\delta^{18}\text{O}$ ratios, confirms an increase in moisture on a regional scale (Bar-Matthews et al., 1999; Ünal-İmer et al., 2015). However, a key finding of our analysis is the variability in the timing of the early Holocene arboreal pollen maximum. While Seki-Eceler recorded its maximum at around 8600 yr BP, sites such as Lakes Gölhisar and Eski Acıgöl show peaks at 9300 and 8000 yr BP, respectively (Eastwood et al., 1999; Roberts et al., 2001). However, the observed asynchrony in the timing of arboreal pollen expansion (9300 yr BP at Lake Gölhisar, 8600 yr BP at Seki-Eceler, and 8000 yr BP at Lake Eski Acıgöl) cannot be attributed solely to 'local moisture variability'. Instead, it probably reflects a combination of chronological uncertainties relating to different dating materials (e.g. bulk sediment versus terrestrial macrofossils), and or basin-specific hydrological thresholds controlling lake levels and vegetation. Finally, the 8.2 ka event emerges as a clear, synchronous regional signal. Its presence in our core, characterised by a period of drought and cold, is supported by multiple records across the region, including pollen data from Dim Cave (Rowe et al., 2020), Lake Nar (England et al., 2008) and $\delta^{18}\text{O}$ records from Central Anatolia (Dean et al., 2015) and Lake Sünnet (Ocakoglu et al., 2013). This widespread evidence shows that the event had a significant and consistent impact on the climate in Anatolia, preventing the growth of forests. Nevertheless, the apparent synchronisation of the 8.2 ka event should be considered approximate, given that the age control and temporal resolution vary considerably among sedimentary, lacustrine, and speleothem archives. Fig. 10 is therefore primarily used to compare the relative timing, direction and magnitude of environmental change across archives, rather than implying perfect chronological equivalence.

Our record from the Seki Basin reveals a significant pattern of climatic instability during the mid-Holocene, which is consistent with regional trends. The period between 8200 and 6750 yr BP, characterised by persistent humidity oscillations and a lack of forest development, is not an isolated phenomenon. Similar oscillations have been observed in other regional records, such as those from Lakes Gölhisar (8800–5100 yr BP; Eastwood et al., 2007) and Eski Acıgöl (8000–6500 yr BP; Roberts et al., 2001), which corroborates this instability. However, these similarities mask important differences in the expression of hydroclimatic variability, which are strongly influenced by basin-specific sensitivity. For example, closed-basin lakes such as Lake Eski Acıgöl may respond rapidly to changes in the evaporation–precipitation balance, whereas larger or hydrologically buffered systems may dampen short-term variability. Additionally, pollen records integrate vegetation responses over longer timescales, resulting in the smoothing of rapid climatic oscillations.

Stable isotope data from Soreq Cave also show slight fluctuations between 8.5 and 5 ka BP (Roberts et al., 2011b) and further support the conclusion that this was a widespread period of fluctuating aridity and humidity across the eastern Mediterranean. As these archives use different dating methods and respond to climate forcing at different timescales, the observed similarities are best interpreted as regionally coherent hydroclimatic tendencies rather than strictly synchronous oscillations. However, a notable point of synthesis lies in the timing of the Mid-Holocene Climatic Optimum. While the Seki region experienced its optimum at around 6000 yr BP, this was not synchronous across all sites. Other records show a significant temporal offset, with the optimum occurring earlier at Lakes Hazar (7750 yr BP; Biltekin et al., 2018) and Gölhisar (8250 yr BP; Eastwood et al., 2007), and later at Lake Eski Acıgöl (5250 yr BP; Roberts et al., 2011b). This suggests that, although the overall climatic improvement of the Mid-Holocene was a regional phenomenon, its local manifestation and timing were influenced by factors such as altitude, moisture availability and local ecological conditions. The 8.2 ka signal is clearly recorded in our core. Our age–depth model places the event between approximately 8.3 and 8.10 ka BP, which is in good agreement with the established chronology for this

climatic downturn. When comparing our record to those from Lakes Gölhisar and Eski Acıgöl, we observe that the onset of the event in our core is broadly synchronous with the signals reported from both Lakes Gölhisar and Eski Acıgöl. The published records for Gölhisar place the event slightly earlier, whereas Lake Eski Acıgöl's record is highly detailed and places the main arid phase within a similar temporal interval. During this time, the TOC and Ca/Ti percentages were high, indicating increasing precipitation and climatic amelioration. At approximately 5500 yr BP, a drought occurred, which caused the TOC content to decrease. Similarly, the AP ratios of Lake Eski Acıgöl and Lake Gölhisar also reflect this event, with a slight decrease in AP at 5800 yr BP (Roberts et al., 2011b). Lake Hazar also experienced a slight decrease in AP at this time, and Ön et al. (2018) reported that humidity gradually increased here until it reached modern values between 5500 and 5000 yr BP. Although the 8.2 ka event is one of the most reliable reference points for inter-archive comparison, precise alignment is still limited due to chronological uncertainty in the Seki-Eceler sequence. We therefore use this comparison to demonstrate the consistency of a major regional climatic perturbation, rather than claiming exact synchronisation with absolute-dated speleothem records. An example of regional asynchrony is the timing of the Mid-Holocene Climatic Optimum. While the Seki-Eceler record indicates optimal conditions at around 6000 yr BP, earlier optima are recorded at Lakes Hazar (around 7750 yr BP) and Gölhisar (around 8250 yr BP), as well as a later optimum at Lake Eski Acıgöl (5250 yr BP). The Beyşehir Occupation Phase (BOP) is a prime example of regional variability. The longer periods were recorded at Lakes Beyşehir, Gölhisar and Söğüt (Eastwood et al., 1998, 1999; Vermeere et al., 2000; Kaniewski et al., 2008). Although the BOP indicates a significant period of human–environment interaction, the way it is reflected in our records suggests that the timing and extent of human influence were not consistent throughout southwestern Anatolia. Our data also highlight significant regional climatic disparities during known global events (Fig. 10). For instance, during the Roman Warm Period (1750–1500 yr BP), the Seki-Eceler record shows a low arboreal pollen (AP) ratio, contrasting with the high AP ratios observed at Lakes Gölhisar, Lake Eski Acıgöl (Roberts et al., 2001) and Hazar (Biltekin et al., 2018), and an increase in $\delta^{18}\text{O}$ in Dim Cave related to the wet climatic period (Baykara et al., 2026). This divergence suggests that, while central Anatolia experienced warmer, wetter conditions conducive to forest growth, the Seki Basin may have remained drier, or may have been subject to other local factors that inhibited forest expansion. This pattern is consistent with the positive $\delta^{18}\text{O}$ trajectories in records from Lake Nar (England et al., 2008) and Soreq Cave (Bar-Matthews et al., 1999), which also indicate a dry period. Similarly, the Dark Ages Cold Period (1500–1150 yr BP) shows a contrasting pattern in our records (Fig. 10). While other regional sites such as Lakes Eski Acıgöl, Gölhisar and Hazar show a decline in AP, indicative of cooling and drying (Roberts et al., 2001; Eastwood et al., 1999; Biltekin et al., 2018). However, the Seki-Eceler record shows a slight increase in AP. This suggests that the climate in our specific locality was relatively warmer or wetter during this period. These contrasts are also evident in the timing of the Little Ice Age (LIA). Our record places the LIA at 650–350 yr BP, which differs from the chronologies reported for Lakes Gölhisar, Hazar, Gölcük, and Dim Cave (Eastwood et al., 1999; Biltekin et al., 2018; Danladi et al., 2023; Baykara et al., 2026). This discrepancy suggests that the local expression and timing of this global climatic event were not synchronised across the broader region. The subrecent period (110 yr BP–present) also exhibits both similarities and differences. The increase in tree taxa around Seki-Eceler, for example, is consistent with findings from Lakes Eski Acıgöl and Hazar (Biltekin et al., 2018), but contrasts with findings from Gölhisar (Eastwood et al., 1999). This demonstrates that even in recent history, local factors continue to play a significant role in shaping vegetation dynamics.

5. Conclusions

Multiproxy palaeoenvironmental analysis of a 4.7 m long core from the archaeological settlement site of the Seki-Eceler Höyük Lycian highlands of southwestern Türkiye has provided valuable insights into the vegetation and climate of southwestern Türkiye from the end of the Pleistocene, throughout the Holocene and up to the present day, a period of approximately 12,000 yr BP.

The key findings that can be drawn from these data are as follows:

- The Seki-Eceler record documents a significant long-term environmental shift, from a cold and dry steppe landscape during the latest Younger Dryas period, to a more humid, productive Early Holocene environment characterised by woodland expansion, marsh formation, and increased catchment activity. During the middle Holocene, forest expansion occurred in the region.
- The multiproxy dataset clearly captures the 8.2 ka event, showing a marked decline in arboreal pollen and an expansion of arid-steppe taxa, as well as geochemical evidence of reduced moisture availability and lower lacustrine productivity.
- The Mid-Holocene Climatic Optimum in the Seki Basin is characterised by increased arboreal pollen, higher plant diversity and geochemical indicators of greater effective moisture and ecological stability. However, comparison with other Anatolian sites shows that the timing of this optimum was not synchronous across the region. This likely reflects a combination of chronological differences between archives, basin-specific hydrological sensitivity and local ecological thresholds.
- The Beyşehir Occupation Phase is expressed weakly and atypically at Seki-Eceler. Although there is evidence of anthropogenic disturbance, the classical OJCV arboricultural signature is not fully developed. The absence of *Juglans* and the sporadic, low representation of *Castanea* are better explained by local edaphic constraints, especially calcareous, clay-rich, high-pH soils than by altitude alone. This suggests that the BOP was not uniformly expressed across southwestern Anatolia, and that local environmental conditions had a strong influence on regional agricultural choices.

CRediT authorship contribution statement

Zahra Hashemi: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Demet Biltekin:** Writing – review & editing, Supervision, Methodology, Investigation. **Alan M. Greaves:** Writing – review & editing, Investigation, Funding acquisition. **Sena Akçer-Ön:** Writing – review & editing, Investigation, Funding acquisition. **Zeki Bora Ön:** Writing – review & editing, Methodology, Investigation, Funding acquisition. **Memet Namık Çağatay:** Writing – review & editing, Methodology, Investigation. **Ömer Yetemen:** Writing – review & editing, Funding acquisition.

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Declaration of competing interest

There are no conflicts of interest associated with the results and/or discussion reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.palaeo.2026.114053>.

Data availability

Supplementary file

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